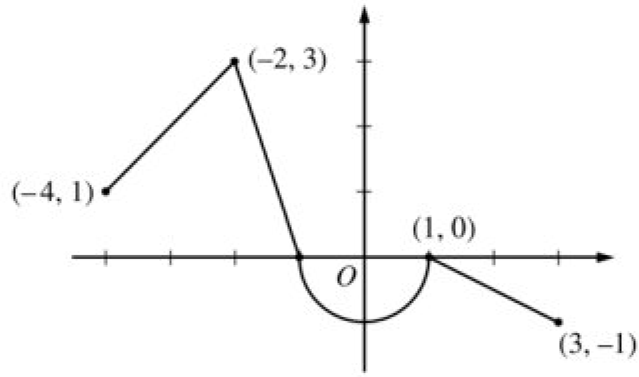
5-4b

1.



Graph of f

Let f be the continuous function defined on [-4, 3] whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g\left(x\right)=\int_1^x f\left(t\right)dt$.

(a) Find the values of $g(2)$ and $g(-2)$.

(b) For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.

(c) Find the x-coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.

(d) For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part A

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2)=\int_1^2 f(t)dt=-\frac{1}{2}(1)\left(\tfrac{1}{2}\right)=-\tfrac{1}{4}$$

$$g(-2)=\int_1^{-2} f(t)dt=-\int_{-2}^{1} f(t)dt$$

$$=-\left(\tfrac{3}{2}-\tfrac{\pi}{2}\right)=\tfrac{\pi}{2}-\tfrac{3}{2}$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for $g(2)$

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1 point is earned for g(-2)

$$g(2) = \int_1^2 f(t)dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$$
$$g(-2) = \int_1^{-2} f(t)dt = -\int_{-2}^{1} f(t)dt$$
$$= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \frac{\pi}{2} - \frac{3}{2}$$

Part B

1 point is earned for g'(-3)

1 point is earned for g''(-3)

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$
$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

0	1	2

The student response earns all of the following points:

1 point is earned for g'(-3)

1 point is earned for g''(-3)

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$
$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

Part C

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

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$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

0	1	2	3

The student response earns all of the following points:

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

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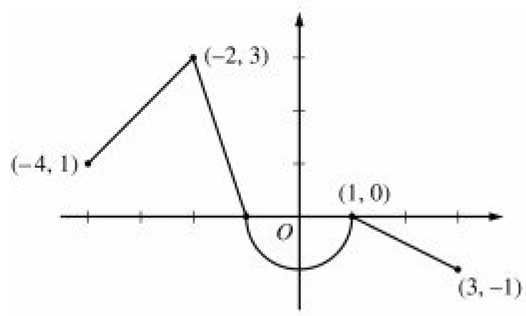
2.



Graph of f

Let f be the continuous function defined on [-4, 3] whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t)dt.$

Find the x-coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.

Part C

One point is earned for considers $g'(x) = 0$

One point is earned for $x = -1$ and $x = 1$

One point is earned for answers with justifications

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

0	1	2	3

The student earns all of the following points:

One point is earned for considers $g'(x) = 0$

One point is earned for $x = -1$ and $x = 1$

One point is earned for answers with justifications

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

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3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The function f is defined by $f(x) = \int_0^x \left(t^2 - 3t - 10\right) dt$.

(a) Write an equation for the line tangent to the graph of f at $x = 6$.

(b) Find all values of x at which f has a critical point. Determine whether f has a relative minimum, a relative maximum, or neither at each of these values. Justify your answers.

(c) On what open intervals, if any, is the graph of f concave down? Give a reason for your answer.

(d) Let g be the function defined by $g(x) = f(\tan x)$. Find $g'\left(\frac{\pi}{4}\right)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An unsupported value of -42 does not earn the first point; the minimal supporting work required is $72 - 54 - 60$.

The second point can be earned without presenting an explicit expression for $f'(x)$ by correctly finding $f'(6)$ with supporting work.

Eligibility for third point: The response has earned the second point and used $f(x) = \frac{1}{3}x^3 - \frac{3}{2}x^2 - 10x$ to find $f(6)$. In this case the third point is earned for any form of the equation $y = f(6) + 8(x - 6)$ for the declared value of $f(6)$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] $f(6)$
- [ ] $f'(x)$

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- [ ] Equation for tangent line

Solution:

$$f(6) = \int_0^6 \left(t^2 - 3t - 10\right)\, dt = \left(\tfrac{1}{3}t^3 - \tfrac{3}{2}t^2 - 10t\right)\Big|_0^6$$

$$= \frac{1}{3} \cdot 6^3 - \frac{3}{2} \cdot 6^2 - 10 \cdot 6 = 72 - 54 - 60 = -42$$

$$f'(x) = x^2 - 3x - 10$$

$$f'(6) = 6^2 - 3 \cdot 6 - 10 = 36 - 18 - 10 = 8$$

An equation for the tangent line is $y = -42 + 8(x - 6)$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

If any additional critical points are listed or either critical point is missing, the first point is not earned.

If only one of $x = -2$ or $x = 5$ is listed as a critical point and that point is correctly justified as a relative maximum or relative minimum, the response earns one point.

“The slope changes from positive to negative” is not a sufficient justification for the second point. The response must be clear about what slope is changing.

Justifications that appeal only to $f(x)$ changing from increasing to decreasing (or vice versa) are not sufficient for the second point.

A response that imports from part (a) an incorrect $f'(x)$ of the form $ax^2 + bx + c$ (typically, $f'(x) = x^2 - 3x$) may earn the first point for finding the critical points associated with their $f'(x)$. Note: The incorrect $f'(x)$ must be imported from part (a) to be eligible for the first point. Such a response is not eligible for the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Critical points
- [ ] Answers with justification

Solution:

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$f'(x) = x^2 - 3x - 10 = (x-5)(x+2) = 0$
$\Rightarrow x = -2$ and $x = 5$

f has a relative maximum at $x = -2$ because $f'(x)$ changes from positive to negative there.

f has a relative minimum at $x = 5$ because $f'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents $2x - 3 < 0$ (or $2x - 3 = 0$ or $2x - 3 > 0$) earns the first point.

To earn the second point, a response must explain that the graph of f is concave down on $\left(-\infty, \frac{3}{2}\right)$ (or $x < \frac{3}{2}$), because $f''(x) < 0$ on that interval.

The interval may include the right endpoint $x = \frac{3}{2}$.

A response that presents any interval other than $\left(-\infty, \frac{3}{2}\right)$ does not earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Find $f''(x)$ and compare to zero
- ☐ Answer with reason

Solution:

$f''(x) = 2x - 3 < 0 \;\Rightarrow\; x < \frac{3}{2}$

The graph of f is concave down on the interval $\left(-\infty, \frac{3}{2}\right)$ because $f''(x) < 0$ there.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $g'(x) = f'(\tan x) \cdot \left(\sec^2 x\right)$.

The answer does not need to be simplified: $\left(\tan^2\left(\frac{\pi}{4}\right) - 3\tan\left(\frac{\pi}{4}\right) - 10\right)\left(\sec^2\left(\frac{\pi}{4}\right)\right)$ or equivalent earns the second point.

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0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Chain rule
- [ ] Answer

Solution:

$g'(x) = f'(\tan x) \cdot (\sec^2 x) = (\tan^2 x - 3\tan x - 10)(\sec^2 x)$

$g'\left(\frac{\pi}{4}\right) = \left(\tan^2\left(\frac{\pi}{4}\right) - 3\tan\left(\frac{\pi}{4}\right) - 10\right)\left(\sec^2\left(\frac{\pi}{4}\right)\right)$
$= (1 - 3 - 10)(2) = -24$

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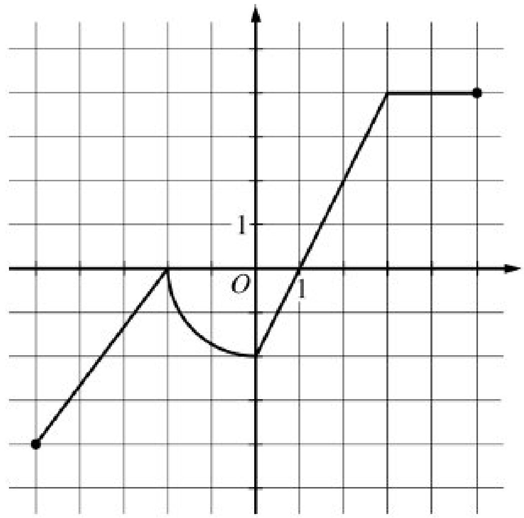
4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.



Graph of f

The graph of the function f, consisting of three line segments and a quarter of a circle, is shown above. Let g be the function defined by $g(x) = \int_1^x f(t)dt$.

(a) Find the average rate of change of g from $x = -5$ to $x = 5$.

(b) Find the instantaneous rate of change of g with respect to x at $x = 3$, or state that it does not exist.

(c) On what open intervals, if any, is the graph of g concave up? Justify your answer.

(d) Find all x-values in the interval $-5 < x < 5$ at which g has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

Part A

1 point is earned for: difference quotient

2 points are earned for: answer

$$\frac{g(5)-g(-5)}{5-(-5)} = \frac{12-(\pi+7)}{10} = \frac{5-\pi}{10}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: difference quotient

2 points are earned for: answer

5-4b

$$\frac{g(5)-g(-5)}{5-(-5)} = \frac{12-(\pi+7)}{10} = \frac{5-\pi}{10}$$

Part B

1 point is earned for: answer

$g'(x) = f(x)$
$g'(3) = f(3) = 4$

The instantaneous rate of change of g at $x = 3$ is 4.

0	1

The student response earns all of the following points:

1 point is earned for: answer

$g'(x) = f(x)$
$g'(3) = f(3) = 4$

The instantaneous rate of change of g at $x = 3$ is 4.

Part C

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.

0	1	2

The student response earns all of the following points:

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.

Part D

1 point is earned for: considers $f(x) = 0$

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1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: considers $f(x) = 0$

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

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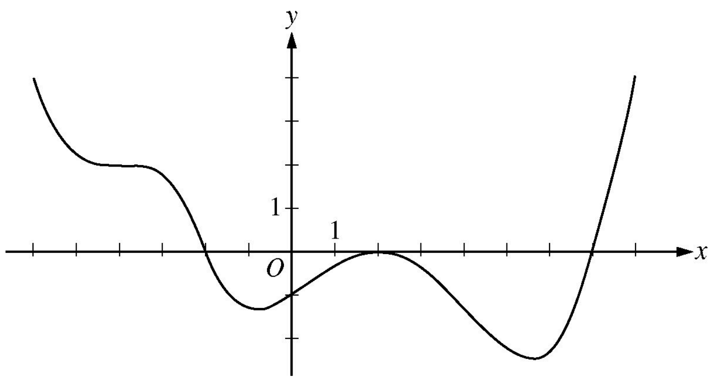
5.



Graph of f'

The figure above shows the graph of f', the derivative of function *f*, for $-6 < x < 8$. Of the following, which best describes the graph of *f* on the same interval?

(A) 1 relative minimum, 1 relative maximum, and 3 points of inflection ✓

(B) 1 relative minimum, 1 relative maximum, and 4 points of inflection

(C) 2 relative minima, 1 relative maximum, and 2 points of inflection

(D) 2 relative minima, 1 relative maximum, and 4 points of inflection

(E) 2 relative minima, 2 relative maxima, and 3 points of inflection

6.

The function *f* is defined on the open interval $0.4 < x < 2.4$ and has first derivative f' given by $f'(x) = \sin(x^2)$. Which of the following statements are true?

i. f has a relative maximum on the interval 0.4 < x < 2.4

ii. f has a relative minimum on the interval 0.4 < x < 2.4

iii. The graph of f has two points of inflection on the interval 0.4 < x < 2.4

(A) I only

(B) II only

(C) III only

(D) I and III only ✓

(E) II and III only

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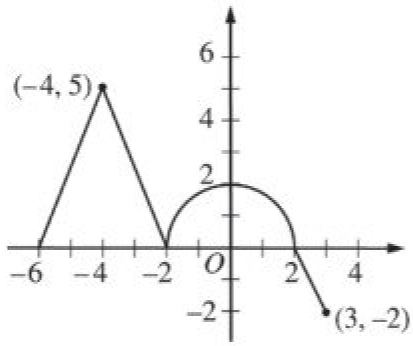
7.



Graph of f

The graph of the continuous function f, consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^{x} f(t)dt$.

Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.

Part C

One point is earned for the horizontal tangent at $x = -2$ and $x = 2$

Two points are earned for the answer with justifications

The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.

0	1	2	3 ✓

The student earns all of the following points:

One point is earned for the horizontal tangent at $x = -2$ and $x = 2$

Two points are earned for the answer with justifications

The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.

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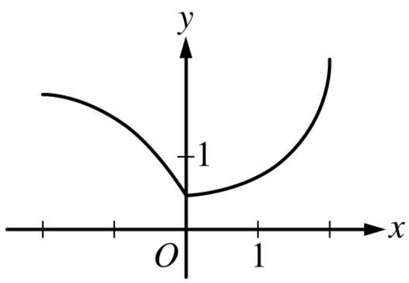
8.



Graph of f

The function f, whose graph is shown above, is defined on the interval $-2 \leq x \leq 2$. Which of the following statements about f is false?

(A) f is continuous at $x = 0$.

(B) f is differentiable at $x = 0$.

(C) f has a critical point at $x = 0$.

(D) f has an absolute minimum at $x = 0$.

(E) The concavity of the graph of f changes at $x = 0$.

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9. Particle X moves along the positive x-axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$
At what time $t \geq 0$ is particle X farthest to the left? Justify your answer.

Part B

The response can earn up to 3 points:

1 point: Consideration of $x'(t) = 0$

1 point: Identification of $t = \frac{6}{5}$

1 point: Correct answer with reason

$$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$

0	1	2	3

The response earns all three of the following points:

1 point: Consideration of $x'(t) = 0$

1 point: Identification of $t = \frac{6}{5}$

1 point: Correct answer with reason

$$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$$

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Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$

10. The function f given by $f(x) = 9x^{2/3} + 3x - 6$ has a relative minimum at $x =$

(A) -8

(B) $-\sqrt[3]{2}$

(C) -1

(D) $-\frac{1}{8}$

(E) 0 ✓

11. The function f has a first derivative given by $f'(x) = x(x-3)^2(x+1)$. At what values of x does f have a relative maximum?

(A) -1 only ✓

(B) 0 only

(C) -1 and 0 only

(D) -1 and 3 only

(E) -1, 0, and 3

12. Let f be the function with first derivative given by $f'(x) = (3 - 2x - x^2)\sin(2x-3)$. How many relative extrema does f have on the open interval $-4 < x < 2$?

(A) Two

(B) Three

(C) Four

(D) Five

(E) Six ✓

13. The derivative of the function f is given by $f'(x) = x^3 - 4\sin(x^2) + 1$. On the interval $(-2.5, 2.5)$, at which of the following values of x does f have a relative maximum?

(A) -1.970 and 0

(B) -1.467 and 1.075

(C) -0.475, 0.542, and 1.396

(D) -0.475 and 1.396 only

(E) 0.542 only ✓

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14. Let *f* be a function whose derivative is given by $f'(x) = \ln\left(x^4 + 5x^3 + x^2 - 7x + 28\right)$. On the open interval $(-4, 1)$, at which of the following values of *x* does *f* attain a relative maximum?

(A) -3.623 only ✓

(B) -0.871 only

(C) -3.623 and -3.284

(D) -3.459 and 0.581 only

(E) -3.459, -0.871 and 0.581

15. The function *f* given by $f(x) = 2x^3 - 3x^2 - 12x$ has a relative minimum at $x =$

(A) -1

(B) 0

(C) 2 ✓

(D) $\frac{3-\sqrt{105}}{4}$

(E) $\frac{3+\sqrt{105}}{4}$

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16. Let f be the function with derivative given by $f'(x) = \sin(x^2 - 3)$. At what values of x in the interval $-3 < x < 3$ does f have a relative maximum?

(A) -1.732 and 2.478 only

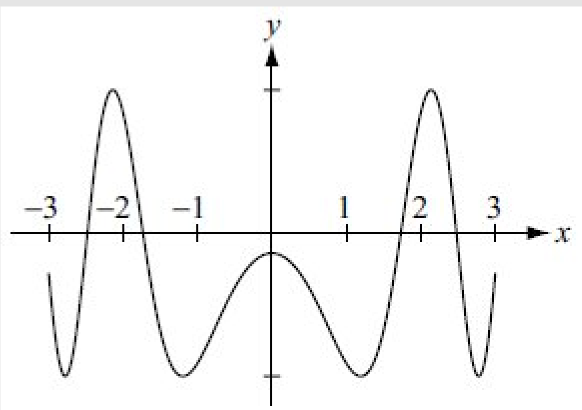
(B) -2.478 and 1.732 only

(C) -2.138, 0, and 2.138

(D) -2.478, -1.732, 1.732, and 2.478

Answer A

This option is correct. The graph of f' on the interval $-3 < x < 3$ shows that the function f will have a relative maximum where the derivative changes from positive to negative . This happens between -2 and -1, and between 2 and 3. The calculator is used to find the zeros of f' on those intervals.



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17. Let g be the function defined by $g(x) = x^4 + 4x^3$. How many relative extrema does g have?

(A) Zero

(B) One ✓

(C) Two

(D) Three

Answer B

This option is correct. First find the critical values where the derivative is 0 or undefined. A relative extrema occurs at a critical value if the derivative changes sign there.

$g'(x) = 4x^3 + 12x^2 = x^2(4x + 12)$

$x^2(4x + 12) = 0$ at $x = 0$ or $x = -3$

There are two zeros, but $g'(x)$ changes sign for $x = -3$ only. The sign of $g'(x)$ at $x = 0$ does not change because of the x^2 factor.

18. If $f\prime(x) = (x-2)(x-3)^2(x-4)^3$, then f has which of the following relative extrema?

I. a relative maximum at $x = 2$

II. a relative maximum at $x = 3$

III. a relative maximum at $x = 4$

(A) I only ✓

(B) III only

(C) I and III only

(D) II and III only

(E) I, II, and III

19. Let f be a function with derivative given by $f'(x) = \frac{x^3-8x^2+3}{\sqrt{x^3+1}}$ for $-1 < x < 9$. At what value of x does f attain a relative maximum?

(A) −0.591

(B) 0

(C) 0.638 ✓

(D) 7.953

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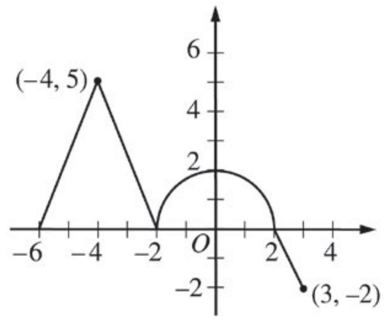
20.



Graph of f

The graph of the continuous function f, consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^{x} f(t)dt$.

Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.

Part C

The response can earn up to 3 points:

1 point: For the horizontal tangent at $x = -2$ and $x = 2$

The graph of g has a horizontal tangent at $x = -2$, $x = 2$, where $g'(x) = f(x) = 0$.

2 points: Correct answer with justifications

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

OR

$g''(x) = f(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = 2$.

0	1	2	3

The response earns all three of the following points:

1 point: For the horizontal tangent at $x = -2$ and $x = 2$

The graph of g has a horizontal tangent at $x = -2$, $x = 2$, where $g'(x) = f(x) = 0$.

2 points: Correct answer with justifications

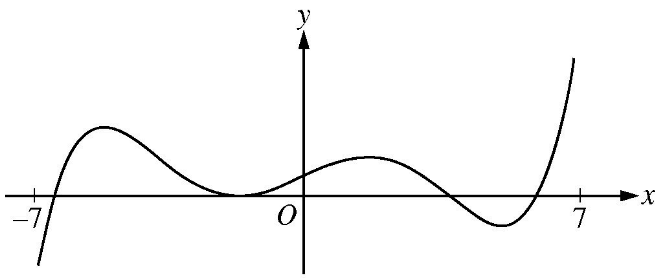
The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

5-4b

OR

$g''(x) = f(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = 2$.

21.



Graph of f'

The figure above shows the graph of f', the derivative of the function f, on the open interval $-7 < x < 7$. If f' has four zeros on $-7 < x < 7$, how many relative maxima does f have on $-7 < x < 7$?

(A) One ✓

(B) Two

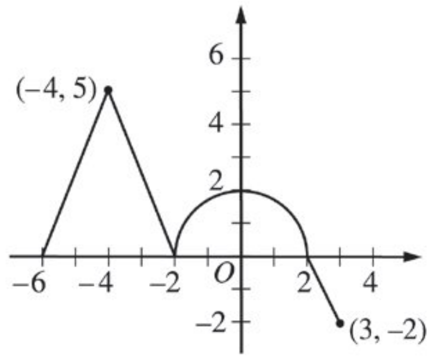
(C) Three

(D) Four

(E) Five

5-4b

22.



Graph of f

The graph of the continuous function f, consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^{x} f(t)dt$.

(a) Find $g(-6)$ and $g(3)$.

(b) Find $g'(0)$.

(c) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.

(d) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.

Part A

The response can earn up to 2 points:

1 point is earned for $g(-6)$

$$g(-6) = \int_{-2}^{-6} f(t)dt = -\int_{-6}^{-2} f(t)dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$$

1 point is earned for $g(3)$

$$g(3) = \int_{-2}^{3} f(t)dt = \frac{1}{2}\pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$

0	1	2

The response earns all of the following points:

1 point is earned for $g(-6)$

5-4b

$$g(-6)=\int_{-2}^{-6} f(t)dt=-\int_{-6}^{-2} f(t)dt=-\frac{1}{2}\cdot 4\cdot 5=-10$$

1 point is earned for $g(3)$

$$g(3)=\int_{-2}^{3} f(t)dt=\frac{1}{2}\pi\cdot 2^2-\frac{1}{2}\cdot 1\cdot 2=2\pi-1$$

Part B

The response can earn up to 1 point:

1 point is earned for the correct evaluation of $g'(0)$

$g'(0) = f(0) = 2$

0	1

The response earns all of the following point:

1 point is earned for the correct evaluation of $g'(0)$

$g'(0) = f(0) = 2$

Part C

The response can earn up to 3 points:

1 point is earned for the horizontal tangent at x = -2 and x = 2

The graph of g has a horizontal tangent at x = -2 and x = 2 where $g'(x) = f(x) = 0$.

2 points are earned for correct answers with justifications

The graph of g has neither a local maximum nor a local minimum at x = -2 because $g'(x) = f(x)$ does not change sign at x = -2.

The graph of g has a local maximum at x = 2 because $g'(x) = f(x)$ changes sign from positive to negative at x = 2.

0	1	2	3

5-4b

The response earns all of the following points:

1 point is earned for the horizontal tangent at $x = -2$ and $x = 2$

The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

2 points are earned for correct answers with justifications

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.

Part D

The response can earn up to 3 points:

2 points are earned for the correct values of x

1 point is earned for the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

0	1	2	3 ✓

The response earns all of the following points:

2 points are earned for the correct values of x

1 point is earned for the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

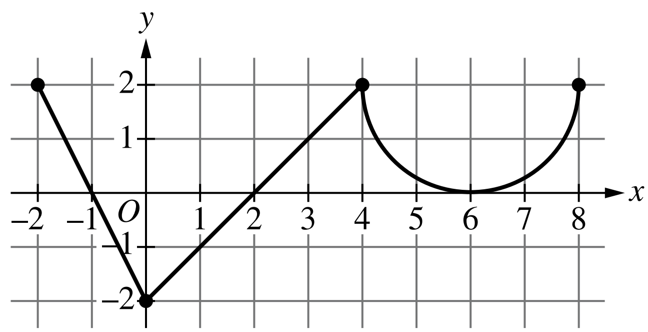
5-4b

23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f'

The function f is defined on the closed interval $[-2, 8]$ and satisfies $f(2) = 1$. The graph of f', the derivative of f, consists of two line segments and a semicircle, as shown in the figure.

(a) Does f have a relative minimum, a relative maximum, or neither at $x = 6$? Give a reason for your answer.

(b) On what open intervals, if any, is the graph of f concave down? Give a reason for your answer.

(c) Find the value of $\lim\limits_{x \to 2} \frac{6f(x) - 3x}{x^2 - 5x + 6}$, or show that it does not exist. Justify your answer.

(d) Find the absolute minimum value of f on the closed interval $[-2, 8]$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that declares $f'(x)$ does not change sign at $x = 6$, so neither, is sufficient to earn the point.

A response does not have to present intervals on which $f'(x)$ is positive or negative, but if any are given, they must be correct. Any presented intervals may include none, one, or both endpoints.

A response that declares $f'(x) > 0$ before and after $x = 6$ does not earn the point.

5-4b

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

$f'(x) > 0$ on $(2, 6)$ and $f'(x) > 0$ on $(6, 8)$.

$f'(x)$ does not change sign at $x = 6$, so there is neither a relative maximum nor a relative minimum at this location.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only by an answer of $(-2, 0)$ and $(4, 6)$, or these intervals including one or both endpoints.

A response must earn the first point to be eligible for second point.

To earn the second point a response must correctly discuss the behavior of f' or the slopes of f'.

Special case: A response that presents exactly one of the two correct intervals with a correct reason earns 1 out of 2 points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Intervals
- [ ] Reason

Solution:

The graph of f is concave down on $(-2, 0)$ and $(4, 6)$ because f' is decreasing on these intervals.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-4b

The first point is earned by the presentation of two separate limits for the numerator and denominator.

A response that presents a limit explicitly equal to $\frac{0}{0}$ does not earn the first point.

The second point is earned by applying L'Hospital's Rule, that is, by presenting at least one correct derivative in the limit of a ratio of derivatives.

The third point is earned for the correct answer with supporting work.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator
- ☐ Uses L'Hospital's Rule
- ☐ Answer

Solution:

Because f is differentiable at $x = 2$, f is continuous at $x = 2$, so $\lim\limits_{x \to 2} f(x) = f(2) = 1$.

$$\lim_{x \to 2} (6f(x) - 3x) = 6 \cdot 1 - 3 \cdot 2 = 0$$

$$\lim_{x \to 2} \left(x^2 - 5x + 6\right) = 0$$

Because $\lim\limits_{x \to 2} \frac{6f(x) - 3x}{x^2 - 5x + 6}$ is of indeterminate form $\frac{0}{0}$, L'Hospital's Rule can be applied.

Using L'Hospital's Rule,
$$\lim_{x \to 2} \frac{6f(x) - 3x}{x^2 - 5x + 6} = \lim_{x \to 2} \frac{6f'(x) - 3}{2x - 5} = \frac{6 \cdot 0 - 3}{2 \cdot 2 - 5} = 3.$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must state $f' = 0$ or equivalent. Listing the zeros of f' is not sufficient.

A response that presents any error in evaluating f at any critical point or endpoint will not earn the justification point.

A response need not present the value of $f(-1)$ provided $x = -1$ is eliminated because it is the location of a local maximum. A response need not present the value of $f(6)$ provided $x = 6$ is eliminated by reference to part (a) or eliminated through analysis.

5-4b

A response need not present the value of $f(8)$ provided there is a presentation that argues $f'(x) \geq 0$ for $x > 2$ and, therefore, $f(8) > f(2)$.

A response that does not consider both endpoints does not earn the justification point.

The answer point is earned only for indicating that the minimum value is 1. It is not earned for noting that the minimum occurs at $x = 2$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Considers $f'(x) = 0$
- [ ] Justification
- [ ] Answer

Solution:

$f'(x) = 0 \Rightarrow x = -1,\, x = 2,\, x = 6$

The function f is continuous on $[-2, 8]$, so the candidates for the location of an absolute minimum for f are $x = -2$, $x = -1$, $x = 2$, $x = 6$, and $x = 8$.

x	$f(x)$
-2	3
-1	4
2	1
6	$7-\pi$
8	$11-2\pi$

The absolute minimum value of f is $f(2) = 1$.

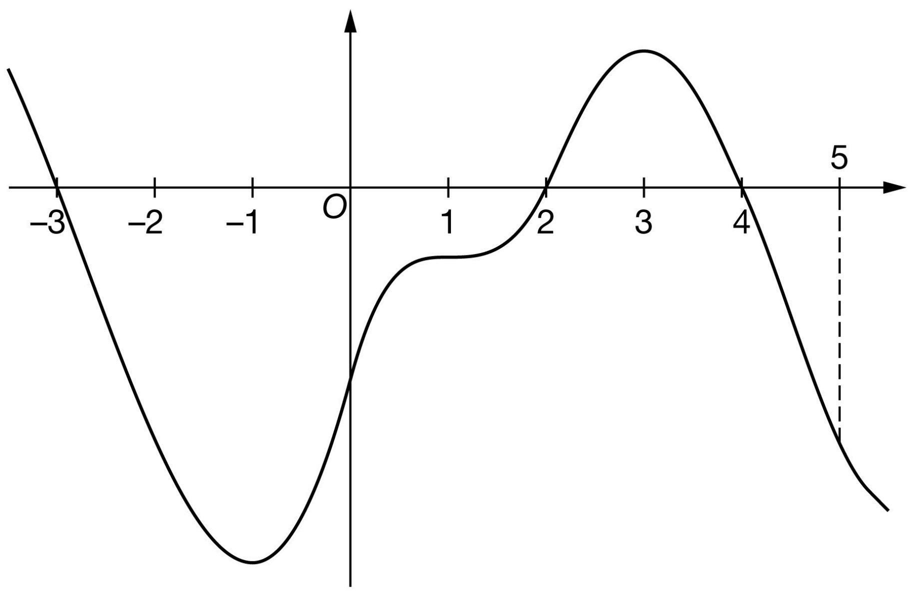
5-4b

24. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The graph of the differentiable function h with domain $-3.5 \le x \le 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of h on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 58, 11, and 8, respectively. The graph of h has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let g be the function defined by $g(x) = \int_4^x h(t)\, dt$ for $-3.5 \le x \le 5.5$.

(a) Find the x-coordinate of each critical point of g on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of g on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $g(5)$.

(e) Find the value of $g(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of b, for $-3 < b < 2$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \le x \le 2$? Justify your answer.

(g) Find $\lim_{x \to 5} \frac{g(x)}{3x}$. Show the computations that lead to your answer.

5-4b

(h) The function a is defined by $a(x) = x^2 \cdot g(x)$. It is known that $h(5) = -16$. Find $a'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3,\ \ x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5,\ \ 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5,\ \ 5.5)$. The point is still earned provided the three critical points, $x = -3,\ \ x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ \ 0),\ \ (2,\ \ 0),\ \ (4,\ \ 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = h(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h, then:

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

5-4b

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$ Any additional error (for example, not justifying that at $x = 2$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-3)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $g(-3.5) < 47$

· $g(-3) = 47$

· $g(2) = -11$

· $g(4) = 0$

· $g(5) = -8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, g has a relative maximum because $g' = h$ is positive for $x < -3$ and $g' = h$ is negative for $x > -3$.

· At $x = 2$, g has a relative minimum because $g' = h$ is negative for $x < 2$ and $g' = h$ is positive for $x > 2$.

· At $x = 4$, g has a relative maximum because $g' = h$ is positive for $x < 4$ and $g' = h$ is negative for $x > 4$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

5-4b

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

At $x = 2$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 4$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only g is eligible for all three points.

If a response gives justifications based only on h, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$.

Justification for the first two points should reference one of the following:

· $g'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $g'(x)$

· $g''(x)$ changes sign

· $h'(x)$ changes sign

“ $g'(x) = h(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to h as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn both points.

5-4b

A response that discusses g' at a particular point (i.e., $g'\left(-3\right)$) when the discussion should have been about $g'\left(x\right)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x=-1$ and $x=3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $x=-1$ with justification
- ☐ $x=3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x=-1$, the graph of g has a point of inflection because $g'\left(x\right)=h\left(x\right)$ changes from decreasing to increasing there.

At $x=3$, the graph of g has a point of inflection because $g'\left(x\right)=h\left(x\right)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -8 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

5-4b

$$g(5) = \int_4^5 h(t)\, dt = -8$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 58 and ± 11 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 58) \pm (\pm 11)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values. The ± 58 and ± 11 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-58) - 11$ earns both points because numerical expressions need not be simplified.

· The expression $(-58) - 11$ earns the first point, but not the second point.

· The expression $-58 - 11 - 8$ earns neither point.

· The expression $\int_4^{-3} h(t)\, dt = 47$ earns neither point.

· The expression $-\int_{-3}^{2} h(t)\, dt - \int_2^4 h(t)\, dt = 47$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 58 and ± 11, but it does earn the second point.

· $-58,\ \ 11$ earns neither point

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\left|\int_{-3}^{2} h(t)\, dt\right| = 58$ and $\left|\int_4^2 h(t) dt\right| = 11$
- ☐ Answer

Solution:

$$g(-3) = \int_4^{-3} h(t)\, dt = -\int_{-3}^{4} h(t)\, dt$$

$$= -\int_{-3}^{2} h(t)\, dt - \int_2^4 h(t)\, dt = -(-58) - 11 = 47$$

5-4b

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3,\ 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $g\ (x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $g\ (x)$ is continuous and differentiable, therefore there must be a value where $g'\ (b)$ equals the average rate of change of $g\ (x)$" earns both points.

· "Yes, $g'\ (x)$ (the derivative of $g\ (x)$) is continuous and differentiable, therefore there must be a value where $g'\ (b)$ equals the average rate of change of $g\ (x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g\ (x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $g\ (x)$ is continuous" earns neither point.

· "Yes, because $g\ (x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = h \Rightarrow g$ is differentiable on $-3 \le x \le 2. \Rightarrow g$ is continuous on $-3 \le x \le 2.$

5-4b

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of b, for $-3 < b < 2$, such that $g'(b)$ equals the average rate of change of g over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\frac{g(5)}{15} = \frac{-8}{15}$
- $\frac{-8}{15}$

Responses that do not earn the point are:

- $\frac{g(5)}{15}$
- $\frac{g(5)}{3 \cdot 5}$

Note: an incorrect value for $g(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g' = h \Rightarrow g$ is continuous. $\Rightarrow \lim_{x\to5} g(x) = g(5)$

$$\lim_{x\to5} \frac{g(x)}{3x} = \frac{g(5)}{3\cdot(5)} = \frac{-8}{15}$$

Part H

5-4b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of a that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or h).

Fundamental Theorem of Calculus is evidenced by linking g' and h either directly or indirectly in the presented derivative of $g(x)$ or in $g'(5)$. Since $h(5) = -16$ is given in the question, replacing $g'(5)$ directly with -16 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g(5)$ that is brought in from part (d) can earn the third point.

$25 \cdot (-16) + 10 \cdot (-8)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a, b, c, or d is incorrect, will earn zero points.

✓

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$a'(x) = x^2 \cdot g'(x) + 2x \cdot g(x)$

$= x^2 \cdot h(x) + 2x \cdot g(x)$

$a'(5) = 25 \cdot h(5) + 2 \cdot 5 \cdot g(5) = 25 \cdot (-16) + 10 \cdot (-8) = -480$

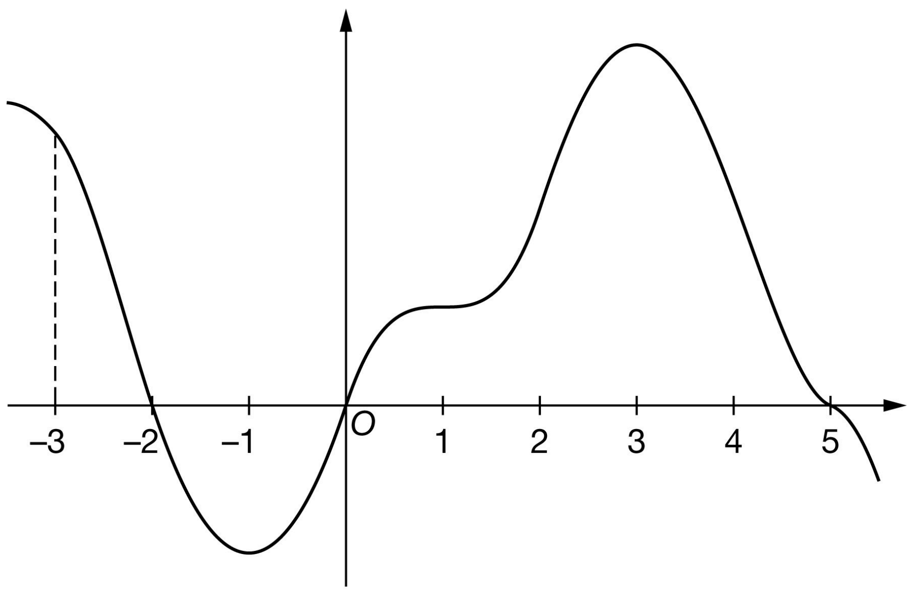
5-4b

25. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-3.5 \le x \le 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-3,-2]$, $[-2,0]$, and $[0,5]$ are 10, 13, and 59, respectively. The graph of g has horizontal tangents at $x=-1$, $x=1$, and $x=3$. Let h be the function defined by $h(x)=\int_0^x g(t)\,dt$ for $-3.5 \le x \le 5.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $h(5)$.

(e) Find the value of $h(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of p, for $-3 < p < 2$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \le x \le 2$? Justify your answer.

(g) Find $\lim\limits_{x \to -3} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

5-4b

(h) The function f is defined by $f(x) = x \cdot h(x)$. It is known that $g(-3) = 18$. Find $f'(-3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -2$, $x = 0$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5,\ 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5,\ 5.5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2,\ 0)$, $(0, 0)$, $(5, 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -2$, $x = 0$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -2$, h has a relative maximum) is

5-4b

eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-2)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-3) = 3$

· $h(-2) = 13$

· $h(0) = 0$

· $h(5) = 59$

· $h(5.5) < 59$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -2$, h has a relative maximum because $h' = g$ is positive for $x < -2$ and $h' = g$ is negative for $x > -2$.

· At $x = 0$, h has a relative minimum because $h' = g$ is negative for $x < 0$ and $h' = g$ is positive for $x > 0$.

· At $x = 5$, h has a relative maximum because $h' = g$ is positive for $x < 5$ and $h' = g$ is negative for $x > 5$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

5-4b

- [ ] Relative maximum with justification
- [ ] Relative minimum with justification
- [ ] Relative maximum with justification

Solution:

At $x = -2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 0$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

Justification for the first two points should reference one of the following:

· $h'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'(x)$

· $h''(x)$ changes sign

· $g'(x)$ changes sign

"$h'(x) = g(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to g as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

5-4b

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 59 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

5-4b

$$h(5) = \int_0^5 g(t)\,dt = 59$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 10 and ± 13 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 10) \pm (\pm 13)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 13 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-10-(-13)$ earns both points because numerical expressions need not be simplified.

· The expression $(-10)-13$ earns the first point, but not the second point.

· The expression $-10-(-13)-3$ earns neither point.

· The expression $\int_0^{-3} g(t)\,dt = 3$ earns neither point.

· The expression $-\int_{-3}^{-2} g(t)\,dt - \int_{-2}^{0} g(t)\,dt = 3$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 10 and ± 13, but it does earn the second point.

· $-10, 13$ earns neither point

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-2}^{-3} g(t)\,dt\right| = 10$ and $\left|\int_{-2}^{0} g(t)\,dt\right| = 13$
- [ ] Answer

Solution:

$$h(-3) = \int_0^{-3} g(t)\,dt = -\int_{-3}^{0} g(t)\,dt$$

$$= -\int_{-3}^{-2} g(t)\,dt - \int_{-2}^{0} g(t)\,dt = -10-(-13) = 3$$

5-4b

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3,\ 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $h\ (x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $h\ (x)$ is continuous and differentiable, therefore there must be a value where $h'\ (p)$ equals the average rate of change of $h\ (x)$ " earns both points.

· "Yes, $h'\ (x)$ (the derivative of $h\ (x)$) is continuous and differentiable, therefore there must be a value where $h'\ (p)$ equals the average rate of change of $h\ (x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $h\ (x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $h\ (x)$ is continuous" earns neither point.

· "Yes, because $h\ (x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 2. \Rightarrow h$ is continuous on $-3 \leq x \leq 2.$

5-4b

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of p, for $-3 < p < 2$, such that $h'(p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\frac{h(-3)}{-6} = \frac{3}{-6}$
- $\frac{3}{-6}$

Responses that do not earn the point are:

- $\frac{h(-3)}{-6}$
- $\frac{h(-3)}{2 \cdot (-3)}$

Note: an incorrect value for $h(-3)$ can be brought in from part (e) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = g \Rightarrow h$ is continuous. $\Rightarrow \lim_{x \to -3} h(x) = h(-3)$

$$\lim_{x \to -3} \frac{h(x)}{2x} = \frac{h(-3)}{2 \cdot (-3)} = \frac{3}{-6} = -\frac{1}{2}$$

Part H

5-4b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of f that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(-3)$. Since $g(-3) = 18$ is given in the question, replacing $h'(-3)$ directly with 18 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(-3)$ that is brought in from part (e) can earn the third point.

$-3 \cdot 18 + 3$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of a, b, or c is incorrect, will earn zero points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$f'(x) = x \cdot h'(x) + 1 \cdot h(x)$$

$$= x \cdot g(x) + h(x)$$

$$f'(-3) = -3 \cdot g(-3) + h(-3) = -3 \cdot (18) + 3 = -51$$

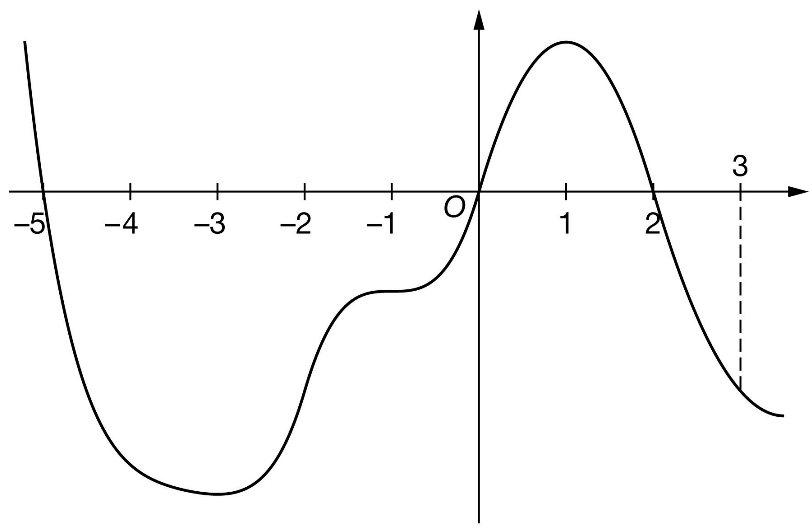
5-4b

26. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-5.5 \leq x \leq 3.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-5, 0]$, $[0, 2]$, and $[2, 3]$ are 57, 12, and 7, respectively. The graph of f has horizontal tangents at $x = -3$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_2^x f(t)\, dt$ for $-5.5 \leq x \leq 3.5$.

(a) Find the x-coordinate of each critical point of g on the interval $-5.5 < x < 3.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of g on the interval $-5.5 < x < 3.5$. Justify your answer.

(d) Find the value of $g(3)$.

(e) Find the value of $g(-5)$. Show the computations that lead to your answer.

(f) Must there exist a value of k, for $-5 < k < 0$, such that $g'(k)$ is equal to the average rate of change of g over the interval $-5 \leq x \leq 0$? Justify your answer.

(g) Find $\lim\limits_{x \to 3} \frac{5g(x)}{2x}$. Show the computations that lead to your answer.

5-4b

(h) The function r is defined by $r(x) = x^2 \cdot g(x)$. It is known that $f(3) = -12$. Find $r'(3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -5, \quad x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5.5,\ 3.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5.5, 3.5)$. The point is still earned provided the three critical points, $x = -5, \quad x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5,\ 0), \quad (0,\ 0), \quad (2,\ 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -5$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

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· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -5$, g has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $f(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., " $g'(-5)$ changes from positive to negative") when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-5.5) < 45$

· $h(-5) = 45$

· $h(0) = -12$

· $h(2) = 0$

· $h(3) = -7$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -5$, g has a relative maximum because $g' = f$ is positive for $x < -5$ and $g' = f$ is negative for $x > -5$.

· At $x = 0$, g has a relative minimum because $g' = f$ is negative for $x < 0$ and $g'(x) = f(x)$ is positive for $x > 0$.

· At $x = 2$, g has a relative maximum because $g' = f$ is positive for $x < 2$ and $g' = f$ is negative for $x > 2$.

0	1	2	3 ✓

5-4b

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -5$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = 0$, g Alt text: g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

At $x = 2$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only g' is eligible for all three points.

If a response gives justifications based only on f, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$.

Justification for the first two points should reference one of the following:

· $g'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $g'(x)$

· $g''(x)$ changes sign

· $f'(x)$ changes sign

“ $g'(x) = f(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to f as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function”

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or "it" or "the slope" with g' produces a response that would earn both points.

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -3$ and $x = 1$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] $x = -3$ with justification
- [ ] $x = 1$ with justification
- [ ] Response includes no other points of inflection

Solution:

At $x = -3$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from decreasing to increasing there.

At $x = 1$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -7 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- [ ] ·Answer

Solution:

5-4b

$$g(3) = \int_2^3 f(t)\ dt = -7$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 57 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 57) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 57 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-57) - (-12)$ earns both points because numerical expressions need not be simplified.

· The expression $(-57) - 12$ earns the first point, but not the second point.

· The expression $-57 - (-12) - 7$ earns neither point.

· The expression $\int_2^{-5} f(t)\ dt = 45$ earns neither point.

· The expression $-\int_{-5}^{0} f(t)dt - \int_0^2 f(t)dt = 45$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 57 and ± 12, but it does earn the second point.

· $-57, -12$ earns neither point

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-5}^{0} f(t)\ dt\right| = 57$ and $\left|\int_2^0 f(t)\ dt\right| = 12$
- [ ] Answer

Solution:

5-4b

$$g(-5) = \int_{2}^{-5} f(t)\, dt = -\int_{-5}^{2} f(t)\, dt$$

$$= -\int_{-5}^{0} f(t)\, dt - \int_{0}^{2} f(t)\, dt = -(-57) - 12 = 45$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-5,\ 0)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $g(x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(k)$ equals the average rate of change of $g(x)$ " earns both points.

· "Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(k)$ equals the average rate of change of $g(x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g(x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $g(x)$ is continuous" earns neither point.

· "Yes, because $g(x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

5-4b

Solution:

$g' = f \Rightarrow g$ is differentiable on $-5 \leq x \leq 0$. $\Rightarrow g$ is continuous on $-5 \leq x \leq 0$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-5 \leq x \leq 0$ to guarantee that there exists a value of k , for $-5 < k < 0$, such that $g'(k)$ equals the average rate of change of g over the interval $-5 \leq x \leq 0$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the first point include:

- $\frac{5(-7)}{2 \cdot 3}$
- $-\frac{35}{6}$

Responses that do not earn the first point include:

- $\frac{5 \cdot g(3)}{6}$
- $\frac{5 \cdot g(3)}{2 \cdot 3}$

Note: an incorrect value for $g(3)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g' = f \Rightarrow g$ is continuous. $\Rightarrow \lim\limits_{x \to 3} g(x) = g(3)$

5-4b

$$\lim_{x \to 3} \frac{5g(x)}{2x} = \frac{5g(3)}{2 \cdot (3)} = \frac{5(-7)}{6} = -\frac{35}{6}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of r that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or f).

Fundamental Theorem of Calculus is evidenced by linking g' and f either directly or indirectly in the presented derivative of $r\,(x)$ or in $r'\,(3)$. Since $f\,(3) = -12$ is given in the question, replacing $g'\,(3)$ directly with -12 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g\,(3)$ that is brought in from part (d) can earn the third point.

$9\,(-12) + 6\,(-7)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a, b, c, or d is incorrect, will earn zero points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Product rule
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$r'\,(x) = x^2 \cdot g'\,(x) + 2x \cdot g\,(x)$

$= x^2 \cdot f\,(x) + 2x \cdot g\,(x)$

$r'\,(3) = 3^2 \cdot f\,(3) + 2 \cdot 3 \cdot g\,(3) = 9 \cdot (-12) + 6 \cdot (-7) = -150$

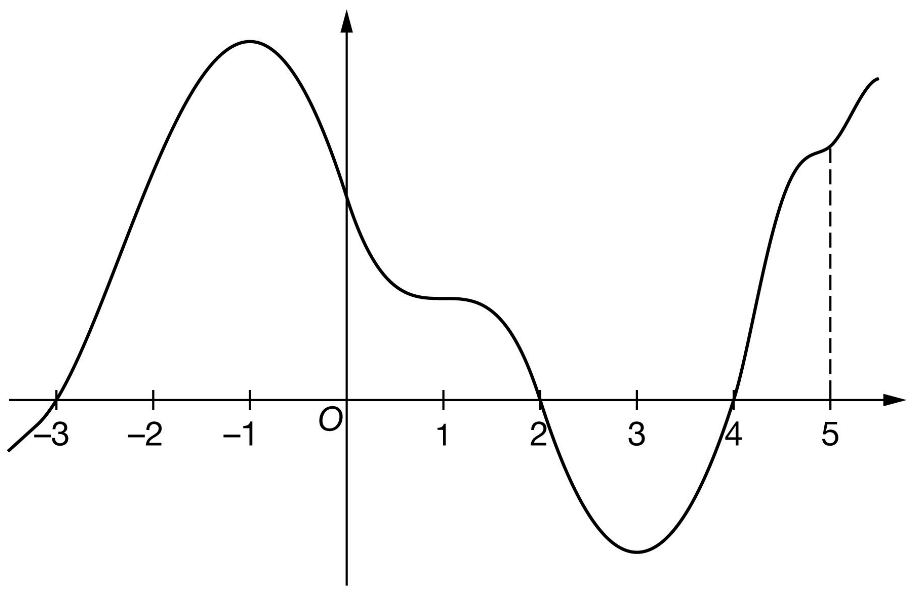
5-4b

27. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 55, 12, and 10, respectively. The graph of g has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let h be the function defined by $h(x) = \int_4^x g(t)\,dt$ for $-3.5 \leq x \leq 5.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $h(5)$.

(e) Find the value of $h(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of r, for $-3 < r < 2$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 2$? Justify your answer.

(g) Find $\lim_{x \to 5} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

5-4b

(h) The function k is defined by $k(x) = x \cdot h(x)$. It is known that $g(5) = 15$. Find $k'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5,\ 5.5)$. The point is still earned provided the three critical points $x = -3$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g.,$(-3, 0)$, $(2, 0)$, $(4, 0)$, read the x -coordinates, and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is

5-4b

eligible for at most one of the three points.

Read references to "the graph" as references to $g(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., "$h'(3)$ changes from negative to positive") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-3.5) > -43$

· $h(-3) = -43$

· $h(2) = 12$

· $h(4) = 0$

· $h(5) = 10$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.

· At $x = 2$, h has a relative maximum because $h' = g$ h prime equals g is positive for $x < 2$ and $h' = g$ is negative for $x > 2$.

· At $x = 4$, h has a relative minimum because $h' = g$ is negative for $x < 4$ and $h' = g$ is positive for $x > 4$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

5-4b

- ☐ Relative minimum with justification
- ☐ Relative maximum with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 4$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

Justification for the first two points should reference one of the following:

· $h'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'(x) = g(x)$

· $h''(x)$ changes sign

· $g'(x)$ changes sign

"$h'(x) = g(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to g as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

5-4b

A response that discusses h' at a particular point (i.e., $h'(-3)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x=-1$ and $x=3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $x=-1$ with justification
- ☐ $x=3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x=-1$, the graph of h has a point of inflection because $h'(x)=g(x)$ changes from increasing to decreasing there.

At $x=3$, the graph of h has a point of inflection because $h'(x)=g(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 10 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

5-4b

$$h(5) = \int_4^5 g(t)\,dt = 10$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 55 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 55) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 55 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-55 - (-12)$ earns both points because numerical expressions need not be simplified.

· The expression $(-55) - 12$ earns the first point, but not the second point.

· The expression $-55 - (-12) - 8$ earns neither point.

· The expression $\int_4^{-3} g(t)\,dt = -43$ earns neither point.

· The expression $-\int_{-3}^{2} g(t)\,dt - \int_2^4 g(t)\,dt = -43$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 55 and ± 12, but it does earn the second point.

· $-55, 12$ earns neither point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $\left|\int_2^{-3} g(t)\,dt\right| = 55$ and $\left|\int_2^4 g(t)\,dt\right| = 12$
- ☐ Answer

Solution:

5-4b

$$h(-3) = \int_{4}^{-3} g(t)\,dt = -\int_{-3}^{4} g(t)\,dt$$

$$= -\int_{-3}^{2} g(t)\,dt - \int_{2}^{4} g(t)\,dt = -55 - (-12) = -43$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $h(x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ " earns both points.

· "Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $h(x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $h(x)$ is continuous" earns neither point.

· "Yes, because $h(x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

5-4b

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 2. \Rightarrow h$ is continuous on $-3 \leq x \leq 2$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of r, for $-3 < r < 2$, such that $h'(r)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\dfrac{h(5)}{10} = 1$
- $\dfrac{10}{10}$

Responses that do not earn the point include:

- $\dfrac{h(5)}{10}$
- 1

Note: an incorrect value for $h(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = g \Rightarrow h$, is continuous. $\Rightarrow \lim\limits_{x \to 5} h(x) = h(5)$

5-4b

$$\lim_{x \to 5} \frac{h(x)}{2x} = \frac{h(5)}{2 \cdot 5} = \frac{10}{10} = 1$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of k that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(5)$. Since $g(5) = 15$ is given in the question, replacing $h'(5)$ directly with 15 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(5)$ that is brought in from part (d) can earn the third point.

$5 \cdot 15 + 10$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of a, b, or c is incorrect, will earn zero points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$k'(x) = x \cdot h'(x) + 1 \cdot h(x)$

$= x \cdot g(x) + h(x)$

$k'(5) = 5g(5) + h(5) = 5 \cdot 15 + 10 = 85$

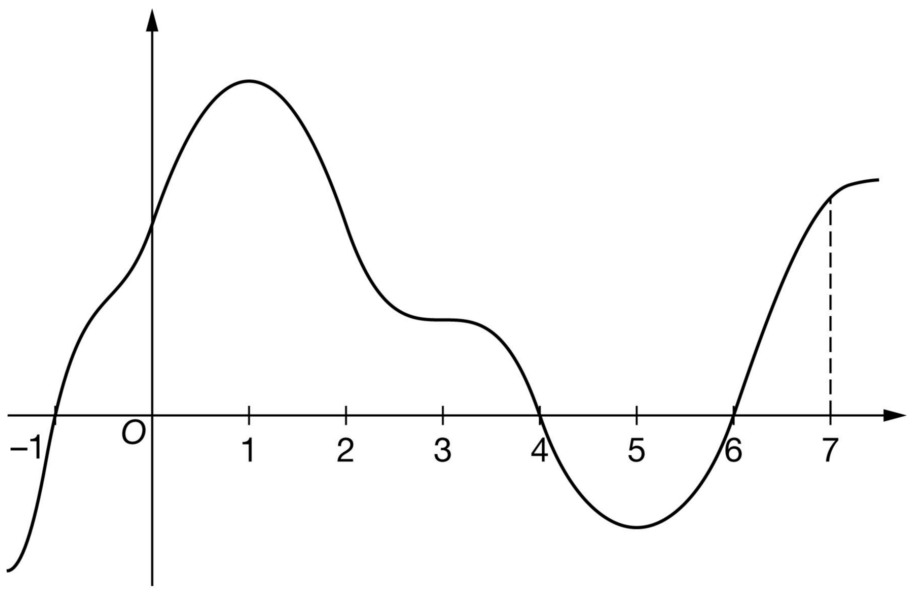
5-4b

28. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-1.5 \le x \le 7.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-1, 4]$, $[4, 6]$, and $[6, 7]$ are 56, 10, and 8, respectively. The graph of f has horizontal tangents at $x = 1$, $x = 3$, and $x = 5$.

Let h be the function defined by $h(x) = \int_6^x f(t)\,dt$ for $-1.5 \le x \le 7.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-1.5 < x < 7.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-1.5 < x < 7.5$. Justify your answer.

(d) Find the value of $h(7)$.

(e) Find the value of $h(-1)$. Show the computations that lead to your answer.

(f) Must there exist a value of d, for $-1 < d < 4$, such that $h'(d)$ is equal to the average rate of change of h over the interval $-1 \le x \le 4$? Justify your answer.

(g) Find $\lim\limits_{x \to 7} \frac{h(x)}{4x}$. Show the computations that lead to your answer.

5-4b

(h) The function q is defined by $q(x) = 3x \cdot h(x)$. It is known that $f(7) = 14$. Find $q'(7)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1.5, 7.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1.5, 7.5)$. The point is still earned provided the three critical points, $x = -1$, $x = 4$, and$x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(4, 0)$, $(6, 0)$, read the x-coordinates and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h\prime(x) = f(x) = 0$ at $x = -1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question. m

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$. Any Any additional error (for example, not justifying that at $x = -1$, h has a relative minimum)

5-4b

is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., "$h'(-1)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-1.5) > -46$

· $h(-1) > -46$

· $h(4) = 10$

· $h(6) = 0$

· $h(7) = 8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -1$, h has a relative minimum because $h' = f$ is negative for $x < -1$ and $h' = f$ is positive for $x > -1$.

· At $x = 4$, h has a relative maximum because $h' = f$ is positive for $x < 4$ and $h' = f$ is negative for $x > 4$.

· At $x = 6$, h has a relative minimum because $h' = f$ is negative for $x < 6$ and $h' = f$ is positive for $x > 6$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

☐ Relative minimum with justification

5-4b

- [ ] Relative maximum with justification
- [ ] Relative minimum with justification

Solution:

At $x = -1$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 4$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

At $x = 6$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on f, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$.

Justification for the first two points should reference one of the following:

· $h'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'(x) = f(x)$

· h'' changes sign

· $f'(x)$ changes sign

" $h'(x) = f(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to f as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be

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penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = 1$ and $x = 5$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $x = 1$ with justification
- ☐ $x = 5$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = 1$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from increasing to decreasing there.

At $x = 5$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 8 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h(7) = \int_6^7 f(t)\, dt = 8$$

5-4b

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 56 and ± 10 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 56) \pm (\pm 10)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 56 and ± 10 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-56 - (-10)$ earns both points because numerical expressions need not be simplified.

· The expression $(-56) - 10$ earns the first point, but not the second point.

· The expression $-56 - (-10) - 8$ earns neither point.

· The expression $\int_6^{-1} g(t)\, dt = -46$ earns neither point.

· The expression $-\int_{-1}^{4} f(t)\, dt - \int_4^6 f(t)\, dt = -46$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 56 and ± 10, but it does earn the second point.

· $-56, \ 10$ earns neither point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\left|\int_4^{-1} f(t)\, dt\right| = 56$ and $\left|\int_4^{6} f(t)\, dt\right| = 10$
- ☐ Answer

Solution:

$$h(-1) = \int_6^{-1} f(t)\, dt = -\int_{-1}^{6} f(t)\, dt$$

$$= -\int_{-1}^{4} f(t)\, dt - \int_4^6 f(t)\, dt = -56 - (-10) = -46$$

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Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-1,\ 4)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· “Yes, $h\ (x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $h\ (x)$ is continuous and differentiable, therefore there must be a value where $h'\ (d)$ equals the average rate of change of $h\ (x)$“ earns both points.

· “Yes, $h'\ (x)$ (the derivative of $h\ (x)$) is continuous and differentiable, therefore there must be a value where $h'\ (d)$ equals the average rate of change of $h\ (x)$“ does not earn the first point, but does the earn second point.

· “Yes because of MVT” does not earn the first point but does earn the second point.

· “Yes, because $h\ (x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $h\ (x)$ is continuous” earns neither point.

· “Yes, because $h\ (x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

$h' = f \Rightarrow h$ is differentiable on $-1 \leq x \leq 4. \Rightarrow h$ is continuous on $-1 \leq x \leq 4.$

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Therefore, the Mean Value Theorem can be applied to h on the interval $-1 \leq x \leq 4$ to guarantee that there exists a value of d, for $-1 < d < 4$, such that $h'(d)$ equals the average rate of change of h over the interval $-1 \leq x \leq 4$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\frac{h(7)}{28} = \frac{2}{7}$
- $\frac{8}{28}$

Responses that do not earn the point include:

- $\frac{h(7)}{4 \cdot 7}$
- $\frac{h(7)}{28}$

Note: an incorrect value for $h(7)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = f \Rightarrow h$ is continuous. $\Rightarrow \lim_{x \to 7} h(x) = h(7)$

$$\lim_{x \to 7} \frac{h(x)}{4x} = \frac{h(7)}{4 \cdot (7)} = \frac{8}{28} = \frac{2}{7}$$

Part H

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of q that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or f).

Fundamental Theorem of Calculus is evidenced by linking h' and f either directly or indirectly in the presented derivative of $q(x)$ or in $q'(7)$. Since $f(7) = 15$ is given in the question, replacing $h'(7)$ directly with 14 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(7)$ that is brought in from part (d) can earn the third point.

$21 \cdot 14 + 3 \cdot 8$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a, b, c, or d is incorrect, will earn zero points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$q'(x) = 3x \cdot h'(x) + 3 \cdot h(x)$

$= 3x \cdot f(x) + 3 \cdot h(x)$

$q'(7) = 3 \cdot 7 \cdot f(7) + 3h(7) = 21 \cdot (14) + 3 \cdot (8) = 318$

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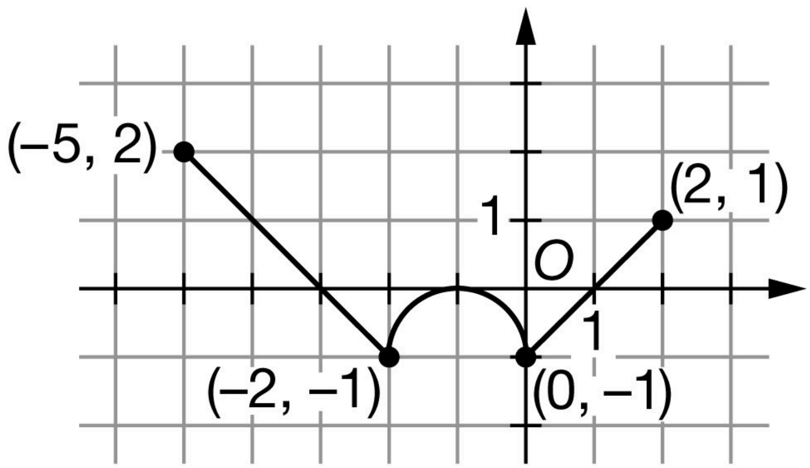
29. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The continuous function g has domain $-5 \leq x \leq 2$. The graph of g, consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = -1$. Let h be the function defined by $h(x) = \int_{-2}^{x} g(t)\,dt$ for $-5 \leq x \leq 2$.

(a) Find the x-coordinate of each critical point of h on the interval $-5 < x < 2$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) For $-5 < x < 2$, on what open intervals is h increasing and concave up? Give a reason for your answer.

(d) Find the value of $h(0)$. Show the computations that lead to your answer.

(e) Find the value of $h(-5)$. Show the computations that lead to your answer.

(f) Find the value of $h''(-3)$, or explain why it does not exist.

(g) Must there exist a value of p, for $-3 < p < 1$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.

(h) Find $\lim_{x \to -2} \frac{h'(x)}{4x}$. Show the computations that lead to your answer.

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(i) The function f is defined by $f(x) = h(-2x)$. Find $f'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3,\ \ x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5,\ 2)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5,\ 2)$. The point is still earned provided the three critical points $x = -3,\ \ x = -1$ and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ \ 0)$, $(-1,\ \ 0)$ and $(1,\ \ 0)$, read the x -coordinates, and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $g(x)$, (that is, $h' = g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h' = g$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is

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eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(1)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3,\ -1,\ 1$ must be included in the table or list of values. The correct values of relevant inputs include:

· $h(-5) = -\frac{3}{2}$

· $h(-3) = \frac{1}{2}$

· $h(-1) = \frac{\pi}{4} - 1$

· $h(1) = \frac{\pi}{2} - \frac{5}{2}$

· $h(2) = \frac{\pi}{2} - 2$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative maximum because $h' = g$ is positive for $x < -3$ and $h' = g$ is negative for $x > -3$.

· At $x = -1$, h has neither because $h' = g$ is negative for $x < -1$ and $h' = g$ is negative for $x > -1$.

· At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

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- [ ] Relative maximum with justification
- [ ] Neither with justification
- [ ] Relative minimum with justification

Solution:

At $x = -3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1,\ 2]$, $(1,\ 2]$, $[1,\ 2)$, and $(1,\ 2)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason " h' is increasing" can also be expressed by either " h'' is positive" or "the slope of h' is positive."

A response referencing just "the function" or "it" or "the slope" is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Answer
- [ ] Reason

Solution:

h is increasing where $h' = g$ is positive.

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h is concave up where $h' = g$ is increasing.

h is increasing and concave up on the open interval $(1,\ 2)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(0) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h(0) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(-5) = -\frac{3}{2}$ with no supporting work earns the second point only.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\int_{-5}^{-3} g(t)\, dt = 2$ – OR – $\int_{-3}^{-2} g(t)\, dt = -\frac{1}{2}$
- [ ] Answer

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Solution:

$$h(-5) = \int_{-2}^{-5} g(t)\, dt = -\int_{-5}^{-2} g(t)\, dt$$

$$= -\int_{-5}^{-3} g(t)\, dt - \int_{-3}^{-2} g(t)\, dt = -(2) - \left(-\tfrac{1}{2}\right) = -\tfrac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(-3) = -1$ or $g'(-3) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-3) = g'(-3) = \frac{2-(-1)}{-5-(-2)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $h(x)$ is continuous and differentiable so MVT applies" earns both points.

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· "Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ " earns both points.

· "Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ " does not earn the first point, but does earn the second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $h(x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $h(x)$ is continuous" earns neither point.

· "Yes, because $h(x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims " h is not differentiable, so therefore the MVT does not apply" earns one of the two points.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 1. \Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of p, for $-3 < p < 1$, such that $h'(p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \to 2} \frac{h'(x)}{4x} = \frac{1}{8}$ earns the point, as does just stating the answer of $\frac{1}{8}$.

No additional supporting work is required.

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The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h' = g \Rightarrow h'$ is continuous. $\Rightarrow \lim_{x \to -2} h'(x) = h'(-2)$

$$\lim_{x \to -2} \frac{h'(x)}{4x} = \frac{h'(-2)}{4 \cdot (-2)} = \frac{g(-2)}{-8} = \frac{-1}{-8} = \frac{1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x, then $x = 1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$\cdot\, h'(-2x) \cdot (-2)$

$\cdot\, g(-2x) \cdot (-2)$

$\cdot\, h'(-2 \cdot 1) \cdot (-2)$

$\cdot\, g(-2 \cdot 1) \cdot (-2)$

Responses that do not earn the point include:

$\cdot\, h'(x) \cdot (-2)$

$\cdot\, g(x) \cdot (-2)$

$\cdot\, h'(-2) \cdot (-2)$

$\cdot\, g(-2) \cdot (-2)$

$\cdot\, h(-2x) \cdot (-2)$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(1)$. For example, the second point is earned by $g(-2x)$ or by $g(-2 \cdot 1)$.

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The second point may be earned without earning the first point. For example, $h'(-2x) = g(-2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -1. The function g does not have to be stated explicitly. For example, $h'(-2 \cdot 1) \cdot (-2) = -1 \cdot (-2)$ earns the first two points. Recall, if $x = 1$ is substituted for x, then $x = 1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $g(-2x) \cdot (-2)$ and $g(-2 \cdot 1) \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of 2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$f'(x) = h'(-2x) \cdot (-2)$

$= g(-2x) \cdot (-2)$

$f'(1) = g(-2 \cdot 1) \cdot (-2) = g(-2) \cdot (-2) = (-1) \cdot (-2) = 2$

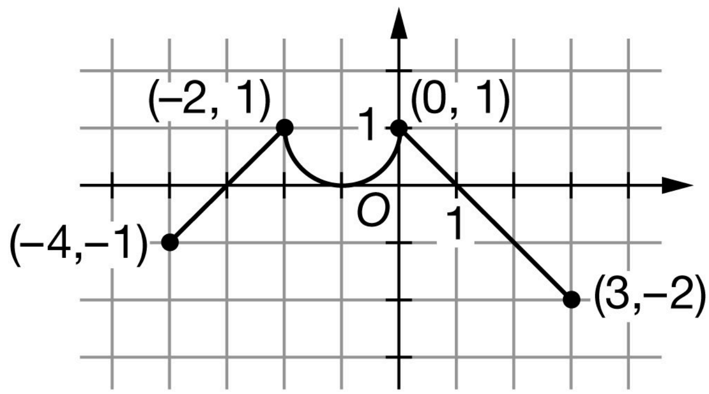
5-4b

30. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The continuous function f has domain $-4 \le x \le 3$. The graph of f, consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = -1$. Let h be the function defined by $h(x) = \int_0^x f(t)\, dt$ for $-4 \le x \le 3$.

(a) Find the x-coordinate of each critical point of h on the interval $-4 < x < 3$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) For $-4 < x < 3$, on what open intervals is h increasing and concave down? Give a reason for your answer.

(d) Find the value of $h(-2)$. Show the computations that lead to your answer.

(e) Find the value of $h(3)$. Show the computations that lead to your answer.

(f) Find the value of $h''(2)$, or explain why it does not exist.

(g) Must there exist a value of k, for $-4 < k < 0$, such that $h'(k)$ is equal to the average rate of change of h over the interval $-4 \le x \le 0$? Justify your answer.

(h) Find $\lim_{x \to -2} \frac{h'(x)}{3x}$. Show the computations that lead to your answer.

(i) The function r is defined by $r(x) = h(-3x)$. Find $r'(-1)$. Show the computations that lead to your answer.

Part A

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3, \ \ x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, \ 3)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, \ 3)$. The point is still earned provided the three critical points $x = -3, \ \ x = -1$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, \ \ 0)$, $(-1, \ \ 0)$, and $(1, \ \ 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = f(x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $f(x)$, (that is, $h' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $h' = f$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

5-4b

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., " $h'(-3)$ changes from negative to positive") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3, \ -1, \ 1$ must be included in the table or list of values. The correct values of relevant inputs include:

· $h(-4) = \frac{\pi}{2} - 2$

· $h(-3) = \frac{\pi}{2} - \frac{5}{2}$

· $h(-1) = \frac{\pi}{4} - 1$

· $h(1) = \frac{1}{2}$

· $h(3) = -\frac{3}{2}$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.

· At $x = -1$, h has neither because $h' = f$ is positive for $x < -1$ and $h' = f$ is positive for $x > -1$.

· At $x = 1$, h has a relative maximum because $h' = f$ is positive for $x < 1$ and $h' = f$ is negative for $x > 1$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

☐ Relative minimum with justification

☐ Neither with justification

☐ Relative maximum with justification

5-4b

Solution:

At $x=-3$, h has a relative minimum because $h'(x)=f(x)$ changes from negative to positive there.

At $x=-1$, h has neither because $h'(x)=f(x)$ does not change sign there.

At $x=1$, h has a relative maximum because $h'(x)=f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-2, \ -1]$ and $(0, \ 1]$ or $[-2, \ -1)$ and $(0, \ 1)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason " h' is decreasing" can also be expressed by either " h'' is negative" or "the slope of h' is negative."

A response referencing just "the function" or "it" or "the slope" is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h'=f$ is positive.

h is concave down where $h'=f$ is decreasing.

h is increasing and concave down on the open intervals $(-2, \ -1)$ and $(0, \ 1)$.

5-4b

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h\left(-2\right)=\ -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h\left(-2\right)=-\left(2\cdot 1-\tfrac{\pi}{2}\cdot 1^2\right)=-\left(2-\tfrac{\pi}{2}\right)=\tfrac{\pi}{2}-2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2}-2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h\left(3\right)=-\frac{3}{2}$ with no supporting work earns the second point only.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

☐ $\int_0^1 f\left(t\right)\,dt=\frac{1}{2}$ – OR – $\int_1^3 f\left(t\right)\,dt=-2$

☐ Answer

Solution:

5-4b

$h\ (3) = \int_0^3 f\ (t)\ dt$

$= \int_0^1 f\ (t)\ dt + \int_1^3 f\ (t)\ dt = \frac{1}{2} + (-2) = -\frac{3}{2}$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(2) = -1$ or $f'\ (2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h''(x) = f'\ (x)$

$h''(2) = f'\ (2) = \frac{1-(-2)}{0-3} = -1$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· “Yes, $h\ (x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $h\ (x)$ is continuous and differentiable, therefore there must be a value where $h'\ (k)$ equals the average rate of change of $h\ (x)$ ” earns both points.

5-4b

· "Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(k)$ equals the average rate of change of $h(x)$" does not earn the first point, but does earn the second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $h(x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $h(x)$ is continuous" earns neither point.

· "Yes, because $h(x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims " h is not differentiable, so therefore the MVT does not apply" earns one of the two points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = f \Rightarrow h$ is differentiable on $-4 \leq x \leq 0$. $\Rightarrow h$ is continuous on $-4 \leq x \leq 0$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-4 \leq x \leq 0$ to guarantee that there exists a value of k, for $-4 < k < 0$, such that $h'(k)$ equals the average rate of change of h over the interval $-4 \leq x \leq 0$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim\limits_{x \to -2} \frac{h'(x)}{3x} = \frac{-1}{6}$ earns the point, as does just stating the answer of $\frac{-1}{6}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

5-4b

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h' = f \Rightarrow h'$ is continuous. $\Rightarrow \lim_{x \to -2} h'(x) = h'(-2)$

$$\lim_{x \to -2} \frac{h'(x)}{3x} = \frac{h'(-2)}{3 \cdot (-2)} = \frac{f(-2)}{-6} = \frac{1}{-6}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x, then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$\cdot\, h'(-3x) \cdot (-3)$

$\cdot\, f(-3x) \cdot (-3)$

$\cdot\, h'(-3 \cdot -1) \cdot (-3)$

$\cdot\, f(-3 \cdot -1) \cdot (-3)$

Responses that do not earn the point include:

$\cdot\, h'(x) \cdot (-3)$

$\cdot\, f(x) \cdot (-3)$

$\cdot\, h'(3) \cdot (-3)$

$\cdot\, f(3) \cdot (-3)$

$\cdot\, h(-3x) \cdot (-3)$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and f either directly or indirectly in the presented derivative of $r(x)$ or in $r'(-1)$. For example, the second point is earned by $f(-3x)$ or by $f(-3 \cdot -1)$.

5-4b

The second point may be earned without earning the first point. For example, $h'(-3x) = f(-3x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(3)$ to -2. The function f does not have to be stated explicitly. For example, $h'(-3 \cdot -1) \cdot (-3) = -2 \cdot (-3)$ earns the first two points. Recall, if $x = -1$ is substituted for x, then $x = -1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $f(-3x) \cdot (-3)$ and $f(-3 \cdot -1) \cdot (-3)$ earn the first two points.

The third point is earned for the correct answer of 6, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$r'(x) = h'(-3x) \cdot (-3)$

$= f(-3x) \cdot (-3)$

$r'(-1) = f((-3) \cdot (-1)) \cdot (-3) = f(3) \cdot (-3) = -2 \cdot (-3) = 6$

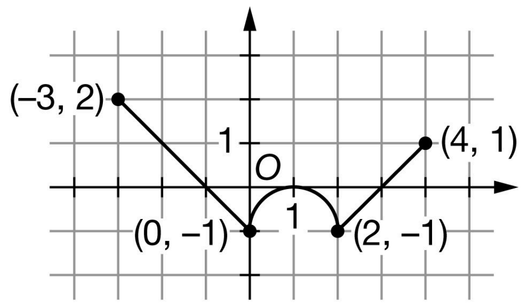
5-4b

31. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The continuous function f has domain $-3 \le x \le 4$. The graph of f, consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = 1$. Let g be the function defined by $g(x) = \int_0^x f(t)\,dt$ for $-3 \le x \le 4$.

(a) Find the x-coordinate of each critical point of g on the interval $-3 < x < 4$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) For $-3 < x < 4$, on what open intervals is g decreasing and concave up? Give a reason for your answer.

(d) Find the value of $g(2)$. Show the computations that lead to your answer.

(e) Find the value of $g(-3)$. Show the computations that lead to your answer.

(f) Find the value of $g''(-2)$, or explain why it does not exist.

(g) Must there exist a value of b, for $-3 < b < 1$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \le x \le 1$? Justify your answer.

(h) Find $\lim_{x \to 2} \frac{g'(x)}{4x}$. Show the computations that lead to your answer.

(i) The function z is defined by $z(x) = g(x^2)$. Find $z'(-2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-4b

To earn this point, the response must include all three critical values: $x = -1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 4)$. The point is still earned provided the three critical points $x = -1$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(3, 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -1$, $x = 1$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $f(x)$, (that is, $g' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $g' = f$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -1$, f has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $f(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

5-4b

A response that discusses g' at a particular point (e.g., " $g'(3)$ changes from negative to positive") when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ values for $x = -1,\ 1,\ 3$ must be included in the table or list of values. The correct values of relevant inputs include:

· $g(-3) = -\frac{3}{2}$

· $g(-1) = \frac{1}{2}$

· $g(1) = \frac{\pi}{4} - 1$

· $g(3) = \frac{\pi}{2} - \frac{5}{2}$

· $g(4) = \frac{\pi}{2} - 2$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -1$, g has a relative maximum because $g' = f$ is positive for $x < -1$ and $g' = f$ is negative for $x > -1$.

· At $x = 1$, g has neither because $g' = f$ is negative for $x < 1$ and $g' = f$ is negative for $x > 1$.

· At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -1$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

5-4b

At $x = 1$, g has neither because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[0,\ 1]$ and $(2,\ 3]$ or $[0,\ 1]$ and $(2,\ 3)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

A response referencing just "the function" or "it" or "the slope" is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

The reason " g' is increasing" can also be expressed by either " g'' is positive" or "the slope of g' is positive."

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where $g' = f$ is negative.

g is concave up where $g' = f$ is increasing.

g is decreasing and concave up on the open intervals $(0,\ 1)$ and $(2,\ 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-4b

Correct supporting work is needed to earn this point.

The correct decimal value is $g\,(2) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

$$g\,(2) = -\left(2 \cdot 1 - \tfrac{\pi}{2} \cdot 1^2\right) = -\left(2 - \tfrac{\pi}{2}\right) = \tfrac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, 2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g\,(-3) = -\frac{3}{2}$ with no supporting work earns the second point only.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\int_{-3}^{-1} f\,(t)\;dt = 2$ – OR – $\int_{-1}^{0} f\,(t)\;dt = -\frac{1}{2}$
- [ ] Answer

Solution:

$$g\,(-3) = \int_{0}^{-3} f\,(t)\;dt = -\int_{-3}^{0} f\,(t)\;dt$$

$$= -\int_{-3}^{-1} f\,(t)\;dt - \int_{-1}^{0} f\,(t)\;dt = -2 - \left(-\tfrac{1}{2}\right) = -\tfrac{3}{2}$$

Part F

5-4b

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(-2) = -1$ or $f'(-2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g''(x) = f'(x)$

$g''(-2) = f'(-2) = \frac{-1-2}{0-(-3)} = -1$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $g\ (x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $g\ (x)$ is continuous and differentiable, therefore there must be a value where $g'\ (b)$ equals the average rate of change of $g\ (x)$ " earns both points.

· "Yes, $g'\ (x)$ (the derivative of $g\ (x)$) is continuous and differentiable, therefore there must be a value where $g'\ (b)$ equals the average rate of change of $g\ (x)$ " does not earn the first point, but does earn the second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g\ (x)$ is continuous and differentiable" earns the first point, but not the second point.

5-4b

· "Yes, because $g(x)$ is continuous" earns neither point.

· "Yes, because $g(x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims " g is not differentiable, so therefore the MVT does not apply" earns one of the two points.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = f \Rightarrow g$ is differentiable on $-3 \le x \le 1$. $\Rightarrow g$ is continuous on $-3 \le x \le 1$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \le x \le 1$ to guarantee that there exists a value of b, for $-3 < b < 1$, such that $g'(b)$ equals the average rate of change of g over the interval $-3 \le x \le 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim\limits_{x\to 2}\frac{g'(x)}{4x} = \frac{-1}{8}$ earns the point, as does just stating the answer of $\frac{-1}{8}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.

0	1

The student response accurately includes the criteria below.

Answer

5-4b

☐

Solution:

$g' = f \Rightarrow g'$ is continuous. $\Rightarrow \lim_{x \to 2} g'(x) = g'(2)$

$\lim_{x \to 2} \frac{g'(x)}{4x} = \frac{g'(2)}{4 \cdot 2} = \frac{f(2)}{8} = \frac{-1}{8}$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -2$ is substituted for x, then $x = -2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

· $g'\left(x^2\right) \cdot 2x$

· $f\left(x^2\right) \cdot 2x$

· $g'\left((-2)^2\right) \cdot 2 \cdot (-2)$

· $f\left((-2)^2\right) \cdot 2 \cdot (-2)$

· $g'(4) \cdot 2 \cdot (-2)$

· $f(4) \cdot 2 \cdot (-2)$

· $g'\left((-2)^2\right) \cdot (-4)$

· $f\left((-2)^2\right) \cdot (-4)$

Responses that do not earn the point include:

· $g'\left(x^2\right) \cdot 2$

· $f\left(x^2\right) \cdot 2$

· $g'(x) \cdot 2x$

· $f(x) \cdot 2x$

· $g'(4) \cdot (-4)$

· $f(4) \cdot (-4)$

5-4b

$\cdot\, g\left(x^2\right) \cdot 2x$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $z(x)$ or in $z'(-2)$. For example, the second point is earned by $f\left(x^2\right)$ or by $f\left((-2)^2\right)$.

The second point may be earned without earning the first point. For example, $g'\left(x^2\right) \cdot 2 = f\left(x^2\right) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

The second point may be earned by going directly from $g'(4)$ to 1. The function f does not have to be stated explicitly. Responses that earn the first two points include:

$\cdot\, g'\left((-2)^2\right) \cdot 2 \cdot (-2) = 1 \cdot 2 \cdot (-2)$

$\cdot\, g'\left((-2)^2\right) \cdot (-4) = 1 \cdot (-4)$

Recall, if $x = -2$ is substituted for x, then $x = -2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f\left(x^2\right) \cdot 2x$ and $f\left((-2)^2\right) \cdot 2 \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of -4, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $g'(x)^2 \cdot 2x$ or $\left(g'(x)\right)^2 \cdot 2x$ are not uncommon. These type of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$z'(x) = g'\left(x^2\right) \cdot 2x$

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$= f\left(x^2\right) \cdot 2x$

$z'(-2) = f\left((-2)^2\right) \cdot 2 \cdot (-2) = f(4) \cdot (-4) = 1 \cdot (-4) = -4$

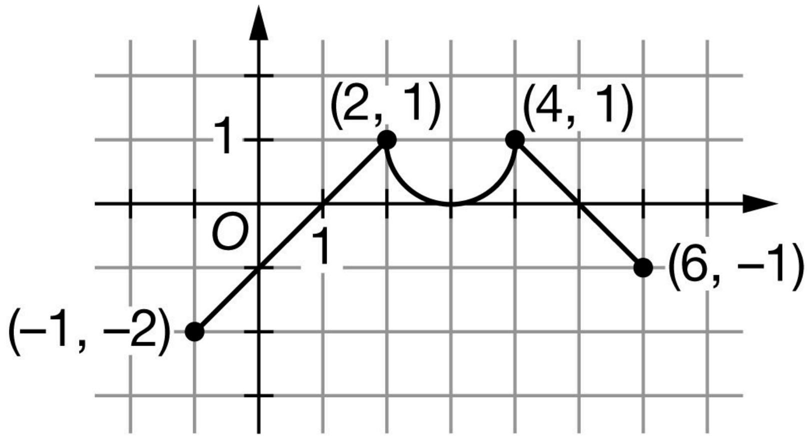
5-4b

32. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The continuous function h has domain $-1 \leq x \leq 6$. The graph of h, consisting of two line segments and a semicircle, is shown in the figure above. The graph of h has a horizontal tangent at $x = 3$. Let g be the function defined by $g(x) = \int_2^x h(t)\, dt$ for $-1 \leq x \leq 6$.

(a) Find the x-coordinate of each critical point of g on the interval $-1 \leq x \leq 6$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) For $-1 \leq x \leq 6$, on what open intervals is g decreasing and concave up? Give a reason for your answer.

(d) Find the value of $g(4)$. Show the computations that lead to your answer.

(e) Find the value of $g(-1)$. Show the computations that lead to your answer.

(f) Find the value of $g''(0)$, or explain why it does not exist.

(g) Must there exist a value of d, for $-1 < d < 3$, such that $g'(d)$ is equal to the average rate of change of g over the interval $-1 \leq x \leq 3$? Justify your answer.

(h) Find $\lim_{x \to 4} \frac{g'(x)}{3x}$. Show the computations that lead to your answer.

(i) The function a is defined by $a(x) = g(2x)$. Find $a'(0)$. Show the computations that lead to your answer.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = 1$, $x = 3$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 6)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 6)$. The point is still earned provided the three critical points $x = 1$, $x = 3$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g.,$(1, 0)$, $(3, 0)$, and $(5, 0)$, read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g'(x) = h(x) = 0$ at $x = 1$, $x = 3$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

· the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $h(x)$, (that is, $g' = h$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $g' = h$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = h(x)$. Any additional error (for example, not justifying that at $x = 1$, g has a relative minimum) is eligible for at most one of the three points.

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Read references to "the graph" as references to $h(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., "$g'(1)$ changes from negative to positive") when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ value for $x = 1, 3, 5$ must be included in the table or list of values. The correct values of relevant inputs include:

· $g(-1) = \frac{3}{2}$

· $g(1) = -\frac{1}{2}$

· $g(3) = 1 - \frac{\pi}{4}$

· $g(5) = \frac{5}{2} - \frac{\pi}{4}$

· $g(6) = 2 - \frac{\pi}{4}$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = 1$, g has a relative minimum because $g' = h$ is negative for $x < 1$ and $g' = h$ is positive for $x > 1$.

· At $x = 3$, g has neither because $g' = h$ is positive for $x < 3$ and $g' = h$ is positive for $x > 3$.

· At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Relative minimum with justification
- [ ] Neither with justification

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- [ ] Relative maximum with justification

Solution:

At $x = 1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 3$, g has neither because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1,\ 1]$, $(-1,\ 1]$, $[-1, 1)$, and $(-1,\ 1)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only h is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason "g' is increasing" can also be expressed by either "g'' is positive" or "the slope of g' is positive."

A response referencing just "the function" or "it" or "the slope" is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Answer
- [ ] Reason

Solution:

g is decreasing where $g' = h$ is negative.

g is concave up where $g' = h$ is increasing.

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g is decreasing and concave up on the open interval $(-1, 1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $g(4) = 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$g(4) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g(-1) = \frac{3}{2}$ with no supporting work earns the second point only.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $\int_{-1}^{1} h(t)\, dt = -2$ – OR – $\int_{1}^{2} h(t)\, dt = \frac{1}{2}$
- ☐ Answer

Solution:

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$$g(-1)=\int_{2}^{-1} h(t)\ dt=-\int_{-1}^{2} h(t)\ dt$$
$$=-\int_{-1}^{1} h(t)\ dt-\int_{1}^{2} h(t)\ dt=-(-2)-\tfrac{1}{2}=\tfrac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(0)=1$ or $h'(0)=1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g''(x)=h'(x)$

$g''(0)=h'(0)=\frac{1}{2}=\frac{1-(-2)}{2-(-1)}$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$ ” earns both points.

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· "Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$" does not earn the first point, but does earn the second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g(x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $g(x)$ is continuous" earns neither point.

· "Yes, because $g(x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims "g is not differentiable, so therefore the MVT does not apply" earns one of the two points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

$g' = h \Rightarrow g$ is differentiable on $-1 \leq x \leq 3. \Rightarrow g$ is continuous on $-1 \leq x \leq 3.$

Therefore, the Mean Value Theorem can be applied to g on the interval $-1 \leq x \leq 3$ to guarantee that there exists a value of d, for $-1 < d < 3$, such that $g'(d)$ equals the average rate of change of g over the interval $-1 \leq x \leq 3.$

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim\limits_{x\to 4} \frac{g'(x)}{3x} = \frac{1}{12}$ earns the point, as does just stating the answer of $\frac{1}{12}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.

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0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g' = h \Rightarrow g'$ is continuous. $\Rightarrow \lim\limits_{x \to 4} g'(x) = g'(4)$

$$\lim_{x \to 4} \frac{g'(x)}{3x} = \frac{g'(4)}{3 \cdot 4} = \frac{h(4)}{12} = \frac{1}{12}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 0$ is substituted for x, then $x = 0$ must appear unsimplified to earn the first point. Responses that earn the point include:

$\cdot\, g'(2x) \cdot 2$

$\cdot\, h(2x) \cdot 2$

$\cdot\, g'(2 \cdot 0) \cdot 2$

$\cdot\, h(2 \cdot 0) \cdot 2$

Responses that do not earn the point include:

$\cdot\, g'(x) \cdot 2$

$\cdot\, h(x) \cdot 2$

$\cdot\, g'(0) \cdot 2$

$\cdot\, h(0) \cdot 2$

$\cdot\, g(2x) \cdot 2$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and h either directly or indirectly in the presented derivative of $a(x)$ or in $a'(0)$. For example, the second point is earned by $h(2x)$ or by $h(2 \cdot 0)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and h are

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linked.

The second point may be earned by going directly from $g'(0)$ to -1. The function h does not have to be stated explicitly. For example, $g'(2 \cdot 0) \cdot 2 = -1 \cdot 2$ earns the first two points. Recall, if $x = 0$ is substituted for x, then $x = 0$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 0) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$a'(x) = g'(2x) \cdot 2$

$= h(2x) \cdot 2$

$a'(0) = g'(2 \cdot 0) \cdot 2 = g'(0) \cdot 2 = h(0) \cdot 2 = -1 \cdot 2 = -2$

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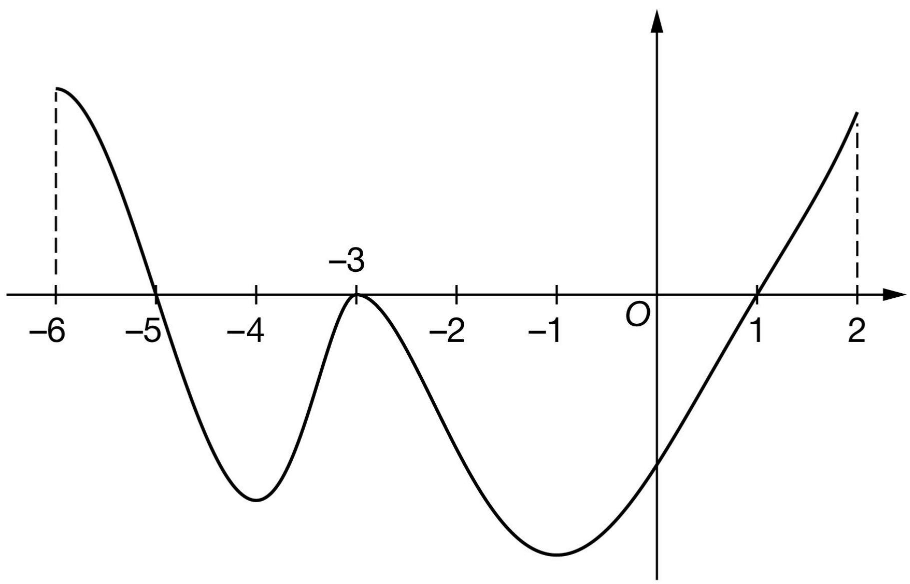
33. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-6 \leq x \leq 2$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-6,-5]$, $[-5,-3]$, $[-3,1]$, and $[1,2]$ are 9, 17, 42, and 6, respectively. The graph of g has horizontal tangents at $x=-4$, $x=-3$, and $x=-1$. Let h be the function defined by $h(x)=\int_{-3}^{x} g(t)\,dt$ for $-6 \leq x \leq 2$.

(a) Find the value of $h(1)$.

(b) Find the value of $h(-6)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of h on the interval $-6 \leq x \leq 2$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(e) For $-6 \leq x \leq 2$, find the x-coordinate of each point of inflection of the graph of h. Justify your answers.

(f) Explain why there must be a value of k, for $-3<k<1$, such that $h(k)=-33$.

(g) Find $\lim\limits_{x \to 1} \frac{3x^2-3}{h(x)+42x}$. Give a reason for your answer.

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(h) The function p is defined by $p(x) = h(2x)$. It is known that $g(-2) = -10.8$. Find $p'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -42 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(1) = \int_{-3}^{1} g(t)\, dt = -42$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 9 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 9 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-9-(-17)$ earns both points because numerical expressions need not be simplified.

· The expression $-9-17$ earns the first point, but not the second point.

· The expression $-9-(-17)+42$ earns neither point.

Note: Just the list $-9, \; -17$ earns neither point. However, if the values -9 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals equal to 8 earns just the second point because not enough work has been shown.

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0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-6}^{-5} f(t)\ dt\right| = 9$ and $\left|\int_{-5}^{-3} f(t)\ dt\right| = 17$
- [ ] Answer

Solution:

$$h(-6) = \int_{-3}^{-6} g(t)\ dt = -\int_{-6}^{-3} g(t)\ dt$$

$$= -\int_{-6}^{-5} g(t)\ dt - \int_{-5}^{-3} g(t)\ dt = -9 - (-17) = 8$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -5$, $x = -3$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-6,\ 2)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-6,\ 2)$. The point is still earned provided the three critical points, $x = -5$, $x = -3$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5,\ 0)$, $(-3,\ 0)$, and $(1,\ 0)$, read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -5$, $x = -3$, and $x = 1$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on $h\prime$ is eligible for all three points.

If a response gives justifications based only on g , then:

· the response is eligible for all three points, if the response presents the connection between $g\left(x\right)$ and $h'\left(x\right)$ (that is, $g\left(x\right)=h'\left(x\right)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g\left(x\right)=h'\left(x\right)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $g\left(x\right)=h'\left(x\right)$. Any additional error (for example, not justifying that at $x=-5$, h has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $g\left(x\right)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., " $h'\left(-5\right)$ changes from positive to negative") when the discussion should have been about $h'\left(x\right)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h\left(x\right)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x=-5$, $x=-3$, $x=1$) must be included in the table or list. The correct values of relevant inputs are:

· $h\left(-6\right)=8$

· $h\left(-5\right)=17$

· $h\left(-3\right)=0$

· $h\left(1\right)=-42$

· $h\left(2\right)=-36$

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Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

· At $x = -5$, h has a relative maximum because $h' = g$ is positive for $x < -5$ and $h' = g$ is negative for $x > -5$.

· At $x = -3$, h has neither because $h\prime= g$ is negative for $x < -3$ and $h' = g$ is negative for $x > -3$.

· At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -3$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-4, \; -3, \; -1$, and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· h' changes from increasing to decreasing or vice versa

· h'' changes from positive to negative or vice versa

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· h' has a local extrema

Any incorrect statement (such as h' changes from increasing to decreasing at $x = -4$) fails to earn the second point.

If the response addresses at least one correct value of x, it is eligible to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Location of inflection points
- [ ] Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -4$, $x = -3$, and $x = -1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow h$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-3 < x < 1$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since h is continuous, and $-42 < -33 < 0$, such a point exists" would earn the point.

The value of -33 must be bounded by -42 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h\left(-3\right)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see A$h\left(-3\right) = 0$.

A response may reference a value of $h\left(1\right)$ determined in part (a). If the value of $h\left(1\right)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -33.

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0	1

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

$h\,(1) = -42 < -33 < 0 = h\,(-3)$

By the Intermediate Value Theorem, there must be a value of k, for $-3 < k < 1$, such that $h\,(k) = -33$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3 \cdot 1^2 - 3 = 0$ and $h\,(1) + 42 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3 \cdot 1^2 = 3$ and $h\,(1) = -42$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim\limits_{x\to1} \frac{3x^2-3}{h\,(x)+42x} = \frac{0}{0}$ does not earn the first point. The statement $\lim\limits_{x\to1} \frac{3x^2-3}{h\,(x)+42x} \to \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x\to1} \frac{3x^2-3}{h\,(x)+42x} = \frac{6x}{h'\,(x)+42}$, will earn the second point, but not the third point.

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Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \to 1} \frac{3x^2-3}{h(x)+42x} = \frac{0}{0}$ and $\lim_{x \to 1} \frac{3x^2-3}{h(x)+42x} = \frac{6x}{h'(x)+42}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{1}{7}$, or its numeric equivalent.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Limits of numerator and denominator equal
- [ ] Application of L'Hospital's Rule
- [ ] Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
$\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \to 1} h(x) = h(1)$ and $\lim_{x \to 1} h'(x) = h'(1)$.

$$\lim_{x \to 1} \left(3x^2 - 3\right) = 3 \cdot 1^2 - 3 = 0$$

$$\lim_{x \to 1} \left(h(x) + 42x\right) = h(1) + 42 \cdot 1 = -42 + 42 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 1} \frac{3x^2-3}{h(x)+42x} = \lim_{x \to 1} \frac{6x}{h'(x)+42}$$

$$= \frac{6 \cdot 1}{h'(1)+42} = \frac{6}{g(1)+42} = \frac{6}{0+42} = \frac{1}{7}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x, then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$\cdot\, h'(2x) \cdot 2$

$\cdot\, g(2x) \cdot 2$

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· $h'(2 \cdot (-1)) \cdot 2$

· $g(2 \cdot (-1)) \cdot 2$

Responses that do not earn the point include:

· $h'(x) \cdot 2$

· $g(x) \cdot 2$

· $h'(-2) \cdot 2$

· $g(-2) \cdot 2$

· $h(2x) \cdot 2$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $p(x)$ or in $p'(-1)$. For example, the second point is demonstrated by $g(2x)$ or by $g(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $h'(2x) = g(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -10.8. The function g does not have to be stated explicitly. For example, $h'(2 \cdot (-1)) \cdot 2 = -10.8 \cdot 2$ earns the first two points. Recall, if $x = -1$ is substituted for x, then $x = -1$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $g(2x) \cdot 2$ and $g(2 \cdot (-1)) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -21.6, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Chain rule
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$p'(x) = h'(2x) \cdot 2$

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$= g\,(2x)\,\cdot\,2$

$p'\,(-1) = g\,(2\,\cdot\,(-1))\,\cdot\,2 = g\,(-2)\,\cdot\,2 = (-10.8)\,\cdot\,2 = -21.6$

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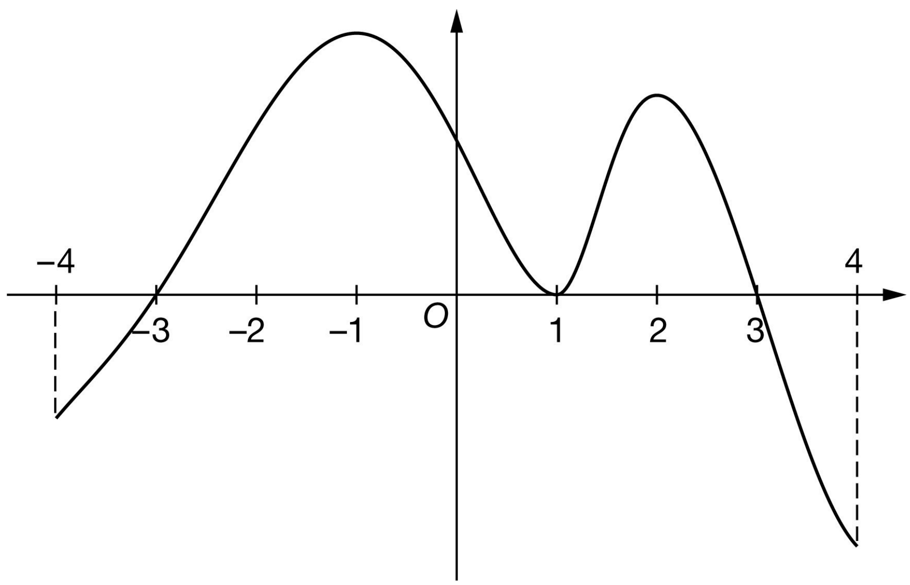
34. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-4 \le x \le 4$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-4, -3]$, $[-3, 1]$, $[1, 3]$, and $[3, 4]$ are 5, 46, 18, and 11, respectively. The graph of f has horizontal tangents at $x = -1$, $x = 1$, and $x = 2$. Let h be the function defined by $h(x) = \int_1^x f(t)\,dt$ for $-4 \le x \le 4$.

(a) Find the value of $h(3)$.

(b) Find the value of $h(-4)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of h on the interval $-4 < x < 4$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(e) For $-4 < x < 4$, find the x-coordinate of each point of inflection of the graph of h. Justify your answers.

(f) Explain why there must be a value of c, for $1 < c < 3$, such that $h(c) = 13$.

(g) Find $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x}$. Give a reason for your answer.

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(h) The function m is defined by $m(x) = h(x^2)$. It is known that $f(4) = -19.3$. Find $m'(2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 18 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(3) = \int_1^3 f(t)\,dt = 18$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 5 and ± 46 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 5 and ± 46 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-5) - 46$ earns both points because numerical expressions need not be simplified.

· The expression $-5 - 46$ earns the first point, but not the second point.

· The expression $-(-5) - 46 + 8$ earns neither point.

Note: Just the list -5, 46 earns neither point. However, if the values -5 and 46 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -41 earns just the second point because not enough work has been shown.

5-4b

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-4}^{-3} f(t)\ dt\right| = 5$ and $\left|\int_{-3}^{1} f(t)\ dt\right| = 46$
- [ ] Answer

Solution:

$$h(-4) = \int_{1}^{-4} f(t)\ dt = -\int_{-4}^{1} f(t)\ dt$$

$$= -\int_{-4}^{-3} f(t)\ dt - \int_{-3}^{1} f(t)\ dt = -(-5) - 46 = -41$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(1, 0)$, and $(3, 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = f(x) = 0$ at $x = -3$, $x = 1$, and $x = 3$.

5-4b

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points, if the response presents the connection between $f(x)$ and $h'(x)$ (that is, $f(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $f(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., "$h'(-3)$ changes from positive to negative") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = 1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

· $h(-4) = -41$

· $h(-3) = -46$

· $h(1) = 0$

· $h(3) = 18$

· $h(4) = 7$

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Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

· At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.

· At $x = 1$, h has neither because $h' = f$ is positive for $x < 1$ and $h' = f$ is positive for $x > 1$.

· At $x = -3$, h has a relative maximum because $h' = f$ is positive for $x < 3$ and $h' = f$ is negative for $x > 3$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 1$, h has neither a relative minimum nor a relative maximum because $h'(x) = f(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-1, 1, 2$ and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· h' changes from increasing to decreasing or vice versa

· h'' changes from positive to negative or vice versa

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· h' has a local extrema

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -1$) fails to earn the second point.

If the response addresses at least one correct value of x it is eligible to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -1$, $x = 1$, and $x = 2$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow$ that h is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since h is continuous, and $0 < 13 < 18$, such a point exists" would earn the point.

The value of 13 must be bounded by 0 and 18 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $h(1) = 0$ must be present.

A response may reference a value of $h(3)$ determined in part (a). If the value of $h(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 13.

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0	1

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$h' = f$ is differentiable on $1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $1 \leq x \leq 3$.

$h(1) = 0 < 13 < 18 = h(3)$

By the Intermediate Value Theorem, there must be a value of c, for $1 < c < 3$, such that $h(c) = 13$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 18 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 18$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{0}{0}$ does not earn the first point. The statement $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} \to \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{2x}{h'(x) - 6}$, will earn the second point, but not the third point.

5-4b

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x\to3}\frac{x^2-9}{h(x)-6x}=\frac{0}{0}$ and $\lim_{x\to3}\frac{x^2-9}{h(x)-6x}=\frac{2x}{h'(x)-6}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of -1, or its numeric equivalent.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Limits of numerator and denominator equal 0
- [ ] Application of L'Hospital's Rule
- [ ] Answer

$h'=f$ is differentiable. $\Rightarrow h'=f$ is continuous.
$\Rightarrow h$ is continuous. $\Rightarrow \lim_{x\to3}h(x)=h(3)$ and $\lim_{x\to3}h'(x)=h'(3)$.

$$\lim_{x\to3}\left(x^2-9\right)=3^2-9=0$$

$$\lim_{x\to3}\left(h(x)-6x\right)=h(3)-6\cdot3=18-18=0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to3}\frac{x^2-9}{h(x)-6x}=\lim_{x\to3}\frac{2x}{h'(x)-6}$$

$$=\frac{2\cdot3}{h'(3)-6}=\frac{6}{f(3)-6}=\frac{6}{0-6}=-1$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x=2$ is substituted for x, then $x=2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$\cdot\, h'(x^2)\cdot2x$

$\cdot\, f(x^2)\cdot2x$

$\cdot\, h'(2^2)\cdot2\cdot2$

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· $f(2^2) \cdot 2 \cdot 2$

· $h'(4) \cdot 2 \cdot 2$

· $f(4) \cdot 2 \cdot 2$

· $h'(2^2) \cdot 4$

· $f(2^2) \cdot 4$

Responses that do not earn the point include:

· $h'(x^2) \cdot 2$

· $f(x^2) \cdot 2$

· $h'(x) \cdot 2x$

· $f(x) \cdot 2x$

· $h'(4) \cdot 4$

· $f(4) \cdot 4$

· $h(x^2) \cdot 2x$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $m(x)$ or in $m'(2)$. For example, the second point is earned by $f(x^2)$ or by $f(2^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = f(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(1)$ to -19.3. The function f does not have to be stated explicitly. Responses that earn the first two points include:

· $h'(2^2) \cdot 2 \cdot 2 = -19.3 \cdot 2 \cdot 2$

· $h'(2^2) \cdot 4 = -19.3 \cdot 4$

Recall, if $x = 2$ is substituted for x, then $x = 2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^2) \cdot 2x$ and $f(2^2) \cdot 2 \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -77.2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

5-4b

Be careful of parenthesis errors in part (h). Responses of the form $h'(x^2) \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$m'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$

$m'(2) = f(2^2) \cdot 2 \cdot 2 = f(4) \cdot 4 = -19.3 \cdot 4 = -77.2$

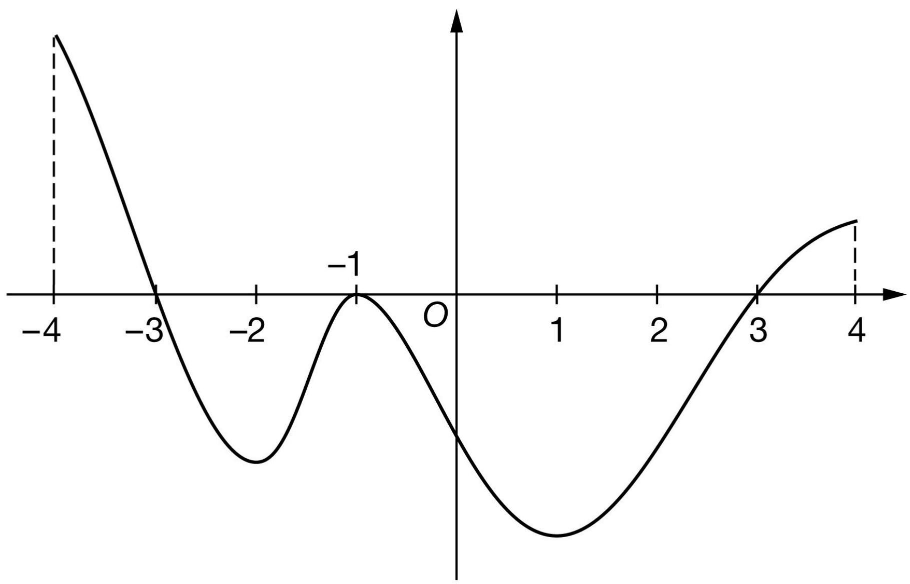
5-4b

35. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-4 \le x \le 4$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 12, 17, 48, and 4, respectively. The graph of f has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_{-1}^{x} f(t)dt$ for $-4 \le x \le 4$.

(a) Find the value of $g(3)$.

(b) Find the value of $g(-4)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of g on the interval $-4 < x < 4$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(e) For $-4 < x < 4$, find the x-coordinate of each point of inflection of the graph of g. Justify your answers.

(f) Explain why there must be a value of d, for $-1 < d < 3$, such that $g(d) = -25$.

(g) Find $\lim\limits_{x \to 3} \frac{x^2 - 9}{g(x) + 16x}$. Give a reason for your answer.

5-4b

(h) The function a is defined by $a(x) = g(x^3)$. It is known that $f(1) = -21.1$. Find $a'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -48 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(3) = \int_{-1}^{3} f(t)\, dt = -48$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 12 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 12 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-12 - (-17)$ earns both points because numerical expressions need not be simplified.

· The expression $-12 - (-17)$ earns the first point, but not the second point.

· The expression $-12 - (-17) + 48$ earns neither point.

Note: Just the list $-12, \ -17$ earns neither point. However, if the values -12 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals 5 earns just the second point because not enough work has been shown.

5-4b

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-3}^{-4} f(t)\ dt\right| = 12$ and $\left|\int_{-3}^{-1} f(t)\ dt\right| = 17$
- [ ] Answer

Solution:

$$g(-4) = \int_{-1}^{-4} f(t)\ dt = -\int_{-4}^{-1} f(t)\ dt$$

$$= -\int_{-4}^{-3} f(t)\ dt - \int_{-3}^{-1} f(t)\ dt = -12 - (-17) = 5$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4,\ 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4,\ 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ 0)$, $(-1,\ 0)$, and $(3,\ 0)$ read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -3$, $x = -1$, and $x = 3$.

5-4b

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points, if the response presents the connection between $f(x)$ and $g'(x)$ (that is, $f(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -3$, g has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $f(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., "$g'(-3)$ changes from positive to negative") when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

· $g(-4) = 5$

· $g(-3) = 17$

· $g(-1) = 0$

· $g(3) = -48$

· $g(4) = -44$

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Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

· At $x = -3$, g has a relative maximum because $g' = f$ is positive for $x < -3$ and $g' = f$ is negative for $x > -3$.

· At $x = -1$, g has neither because $g' = f$ is negative for $x < -1$ and $g' = f$ is negative for $x > -1$.

· At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = -1$, g has neither a relative minimum nor a relative maximum because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-2, \; -1, \; 1$, and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· g' changes from increasing to decreasing or vice versa

· g'' changes from positive to negative or vice versa

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· g' has a local extrema

Any incorrect statement (such as g' changes from increasing to decreasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x, it is eligible to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Location of inflection points
- [ ] Justification

Solution:

The points of inflection for the graph of g occur where $g' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that g is differentiable $\Rightarrow$ g is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since g is continuous, and $-48 < -25 < 0$, such a point exists" would earn the point.

The value of -25 must be bounded by -48 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g(-1) = 0$ must be present.

A response may reference a value of $g(3)$ determined in part (a). If the value of $g(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -25.

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0	1

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$g' = f$ is differentiable on $-1 \le x \le 3$. $\Rightarrow g$ is continuous on $-1 \le x \le 3$.

$g\,(3) = -48 < -25 < 0 = g\,(-1)$

By the Intermediate Value Theorem, there must be a value of d, for $-1 < d < 3$, such that $g\,(d) = -25$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3^2 - 9 = 0$ and $g\,(3) + 48 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $g\,(3) = -48$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim\limits_{x \to 3} \frac{x^2-9}{g(x)+16x} = \frac{0}{0}$ does not earn the first point. The statement $\lim\limits_{x \to 3} \frac{x^2-9}{g(x)+16x} \to \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x \to 3} \frac{x^2-9}{g(x)+16x} = \frac{2x}{g'(x)+16}$, will earn the second point, but not the third point.

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Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x\to 3}\frac{x^2-9}{g(x)+16x}=\frac{0}{0}$ and $\lim_{x\to 3}\frac{x^2-9}{g(x)+16x}=\frac{2x}{g'(x)+16}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{8}$, or its numeric equivalent.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = f$ is differentiable. $\Rightarrow g' = f$ is continuous.
$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x\to 3} g(x) = g(3)$ and $\lim_{x\to 3} g'(x) = g'(3)$.

$$\lim_{x\to 3}\left(x^2-9\right) = 3^2 - 9 = 0$$

$$\lim_{x\to 3}\left(g(x)+16x\right) = g(3) + 16\cdot 3 = -48+48 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to 3}\frac{x^2-9}{g(x)+16x} = \lim_{x\to 3}\frac{2x}{g'(x)+16}$$

$$=\frac{2\cdot 3}{g'(3)+16} = \frac{6}{f(3)+16} = \frac{6}{0+16} = \frac{3}{8}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x=1$ is substituted for x, then $x=1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$\cdot\, g'\left(x^3\right)\cdot 3x^2$

$\cdot\, f\left(x^3\right)\cdot 3x^2$

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$\cdot\, g'\left(1^3\right) \cdot 3 \cdot 1^2$

$\cdot\, f\left(1^3\right) \cdot 3 \cdot 1^2$

$\cdot\, g'(1) \cdot 3 \cdot 1^2$

$\cdot\, f\left(1^3\right) \cdot 3 \cdot 1$

Responses that do not earn the point include:

$\cdot\, g'\left(x^3\right) \cdot 3x$

$\cdot\, f\left(x^3\right) \cdot 3x$

$\cdot\, g'(x) \cdot 3x^2$

$\cdot\, f(x) \cdot 3x^2$

$\cdot\, g'(1) \cdot 3$

$\cdot\, f(1) \cdot 3$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $a(x)$ or in $a'(1)$. For example, the second point is earned by $f\left(x^3\right)$ or by $f\left(1^3\right)$.

The second point may be earned without earning the first point. For example, $g'\left(x^3\right) \cdot 3x = f\left(x^3\right) \cdot 3x$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

The second point may be earned by going directly from $g'(1)$ to -21.1. The function f does not have to be stated explicitly. For example, $g'\left(1^3\right) \cdot 3 \cdot 1^2 = -21.1 \cdot 3 \cdot 1$ earns the first two points. Recall, if $x = 1$ is substituted for x, then $x = 1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f\left(x^3\right) \cdot 3x^2$ and $f\left(1^3\right) \cdot 3 \cdot 1^2$ earn the first two points.

The third point is earned for the correct answer of -63.3, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $g'(x)^3 \cdot 3x^2$ or $\left(g'(x)\right)^3 \cdot 3x^2$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.

0	1	2	3

5-4b

The student response accurately includes **all three** of the criteria below.

- [ ] Chain rule
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$a'(x) = g'\left(x^3\right) \cdot 3x^2$

$= f\left(x^3\right) \cdot 3x^2$

$a'(1) = g'\left(1^3\right) \cdot 3 \cdot 1^2 = g'(1) \cdot 3 = f(1) \cdot 3 = (-21.1) \cdot 3 = -63.3$

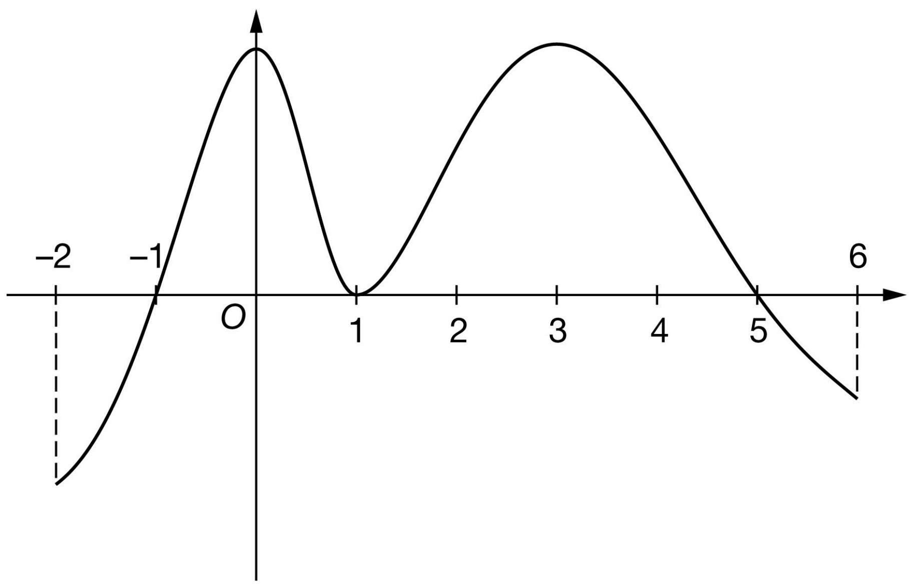
5-4b

36. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of *h*

The graph of the differentiable function h with domain $-2 \leq x \leq 6$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of h on the intervals $[-2, -1]$, $[-1, 1]$, $[1, 5]$, and $[5, 6]$ are 8, 19, 40, and 4, respectively. The graph of h has horizontal tangents at $x = 0$, $x = 1$, and $x = 3$. Let g be the function defined by $g(x) = \int_1^x h(t)\,dt$ for $-2 \leq x \leq 6$.

(a) Find the value of $g(5)$.

(b) Find the value of $g(-2)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of g on the interval $-2 < x < 6$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(e) For $-2 < x < 6$, find the x-coordinate of each point of inflection of the graph of g. Justify your answers.

(f) Explain why there must be a value of w, for $1 < w < 5$, such that $g(w) = 5$.

(g) Find $\lim\limits_{x \to 5} \frac{x^2 - 25}{g(x) - 8x}$. Give a reason for your answer.

(h) The function z is defined by $z(x) = g(2x)$. It is known that $h(4) = 11.25$. Find $z'(2)$. Show the computations that lead to your answer.

5-4b

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 40 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(5) = \int_1^5 h(t)\, dt = 40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 8 and ± 19 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 8 and ± 19 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-8) - 19$ earns both points because numerical expressions need not be simplified.

· The expression $(-8) - 19$ earns the first point, but not the second point.

· The expression $-(-8) - 19 - 40$ earns neither point.

Note: Just the list $-8,\ 19$ earns neither point. However, if the values -8 and 19 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -11 earns just the second point because not enough work has been shown.

5-4b

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\left|\int_{-2}^{-1} h(t)\ dt\right| = 8$ and $\left|\int_{1}^{-1} h(t)\ dt\right| = 19$
- ☐ Answer

Solution:

$$g(-2) = \int_{1}^{-2} h(t)\ dt = -\int_{-2}^{1} h(t)\ dt$$

$$= -\int_{-2}^{-1} h(t)\ dt - \int_{-1}^{1} h(t)\ dt = -(-8) - 19 = -11$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1,$, $x = 1,$,and $x = 5.$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-2,\ 6)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-2,\ 6)$. The point is still earned provided the three critical points, $x = -1$, $x = 1$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1,\ 0)$, $(1,\ 0)$, and $(5,\ 0)$ read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$g'(x) = h(x) = 0$ at $x = -1$, $x = 1$, and $x = 5$.

5-4b

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justification based on g' is eligible for all three points.

If a response gives justifications based only on h, then:

· the response is eligible for all three points, if the response presents the connection between $h\left(x\right)$ and $g'\left(x\right)$ (that is, $h\left(x\right)=g'\left(x\right)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of h or g' as $h\left(x\right)=g'\left(x\right)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $h\left(x\right)=g'\left(x\right)$. Any additional error (for example, not justifying that at $x=-1$, g has a relative minimum) is eligible for at most one of the three points.

Read references to "the graph" as references to $h\left(x\right)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., " $g'\left(-1\right)$ changes from negative to positive") when the discussion should have been about $g'\left(x\right)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g\left(x\right)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x=-1$, $x=1$, $x=5$) must be included in the table or list. The correct values of relevant inputs are:

· $g\left(-2\right)=-11$

· $g\left(-1\right)=-19$

· $g\left(1\right)=0$

· $g\left(5\right)=40$

· $g\left(6\right)=36$

5-4b

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

· At $x = -1$, g has a relative minimum because $g' = h$ is negative for $x < -1$ and $g' = h$ is positive for $x > -1$.

· At $x = 1$, g has neither because $g' = h$ is positive for $x < 1$ and $g' = h$ is positive for $x > 1$.

· At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Relative minimum with justification
- [ ] Neither with justification
- [ ] Relative maximum with justification

Solution:

At $x = -1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 1$, g has neither a relative minimum nor a relative maximum because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $0,\ 1,\ 3$, and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $h(x)$ is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· g' changes from increasing to decreasing or vice versa

· g'' changes from positive to negative or vice versa

5-4b

$\cdot\, g'$ has a local extrema

Any incorrect statement (such as g' changes from decreasing to increasing at $x = 0$) fails to earn the second point.

If the response addresses at least one correct value of x, it is eligible to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Location of inflection points
- [ ] Justification

Solution:

The points of inflection for the graph of g occur where $g' = h$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = 0$, $x = 1$, and $x = 3$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that " g is differentiable $\Rightarrow$ g is continuous."

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 5$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since g is continuous, and $0 < 5 < 40$, such a point exists" would earn the point.

The value of 5 must be bounded by 0 and 40 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g\left(1\right)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g\left(1\right) = 0$ must be present.

A response may reference a value of $g\left(5\right)$ determined in part (a). If the value of $g\left(5\right)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 5.

5-4b

0	1

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$g' = h$ is differentiable on $1 \leq x \leq 5$. $\Rightarrow g$ is continuous on $1 \leq x \leq 5$.

$g(1) = 0 < 5 < 40 = g(5)$

By the Intermediate Value Theorem, there must be a value of w, for $1 < w < 5$, such that $g(w) = 5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $5^2 - 25 = 0$ and $g(5) - 40 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $5^2 = 25$ and $g(5) = 40$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x} = \frac{0}{0}$ does not earn the first point. The statement $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x} \to \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x} = \frac{2x}{g'(x)-8}$, will earn the second point, but not the third point.

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Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim\limits_{x\to 5}\frac{x^2-25}{g(x)-8x}=\frac{0}{0}$ and $\lim\limits_{x\to 5}\frac{x^2-25}{g(x)-8x}=\frac{2x}{g'(x)-8}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-5/4$, or its numeric equivalent.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g'=h$ is differentiable. $\Rightarrow g'=h$ is continuous.
$\Rightarrow g$ is continuous. $\Rightarrow \lim\limits_{x\to 5} g(x)=g(5)$ and $\lim\limits_{x\to 5} g'(x)=g'(5)$.

$$\lim_{x\to 5}\left(x^2-25\right)=5^2-25=0$$

$$\lim_{x\to 5}\left(g(x)-8x\right)=g(5)-8\cdot 5=40-40=0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to 5}\frac{x^2-25}{g(x)-8x}=\lim_{x\to 5}\frac{2x}{g'(x)-8}$$

$$=\frac{2\cdot 5}{g'(5)-8}=\frac{10}{h(5)-8}=\frac{10}{0-8}=-\frac{5}{4}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x=2$ is substituted for x, then $x=2$ must appear unsimplified to earn the first point. Responses that earn the point include:

$\cdot\, g'(2x)\cdot 2$

$\cdot\, h(2x)\cdot 2$

5-4b

· $g'(2 \cdot 2) \cdot 2$

· $h(2 \cdot 2) \cdot 2$

Responses that do not earn the point include:

· $g'(x) \cdot 2$

· $h(x) \cdot 2$

· $g'(2) \cdot 2$

· $h(2) \cdot 2$

· $g(2x) \cdot 2$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking g' prime and h either directly or indirectly in the presented derivative of $z(x)$ or in $z'(2)$. For example, the second point is demonstrated by $h(2x)$ or by $h(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the Chain rule is applied incorrectly, but does earn the second point because g' and h are linked.

The second point may be earned by going directly from $g'(4)$ to 11.25. The function h does not have to be stated explicitly. For example, $g'(2 \cdot 2) \cdot 2 = 11.25 \cdot 2$ earns the first two points. Recall, if $x = 2$ is substituted for x, then $x = 2$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 2) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of 22.5, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$z'(x) = g'(2x) \cdot 2$

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$= h\ (2x)\ \cdot\ 2$

$z'\ (2) = h\ (2\ \cdot\ 2)\ \cdot\ 2 = h\,(4)\ \cdot\ 2 = 11.25\ \cdot\ 2 = 22.5$

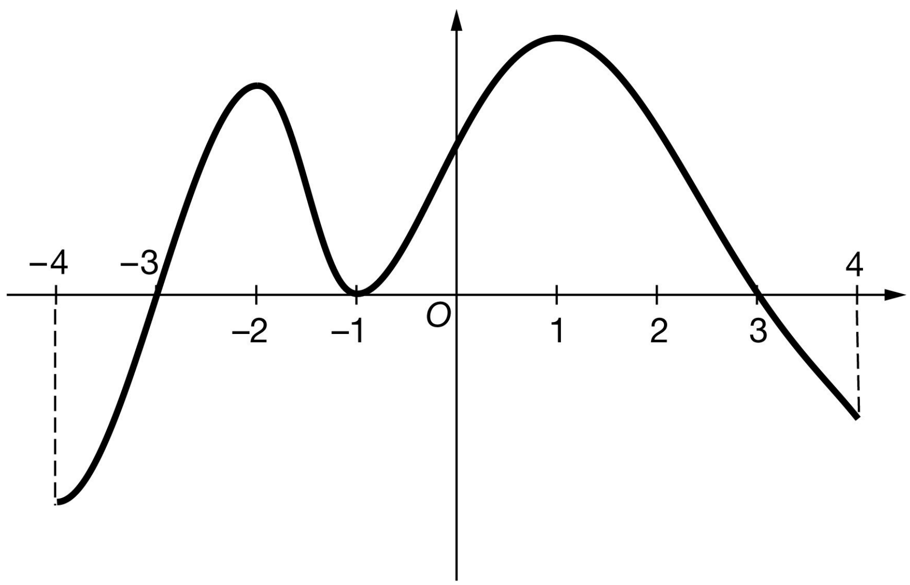
5-4b

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-4,-3]$, $[-3,-1]$, $[-1,3]$, and $[3,4]$ are 10, 18, 45, and 5, respectively. The graph of g has horizontal tangents at $x=-2$, $x=-1$, and $x=1$. Let h be the function defined by $h(x)=\int_{-1}^{x} g(t)\,dt$ for $-4 \leq x \leq 4$.

(a) Find the value of $h(3)$.

(b) Find the value of $h(-4)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of h on the interval $-4<x<4$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(e) For $-4<x<4$, find the x-coordinate of each point of inflection of the graph of h. Justify your answers.

(f) Explain why there must be a value of r, for $-1<r<3$, such that $h(r)=20$.

(g) Find $\lim\limits_{x \to 3} \frac{x^2-9}{h(x)-15x}$. Give a reason for your answer.

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(h) The function k is defined by $k(x) = h(x^2)$. It is known that $g(1) = 19.7$. Find $k'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 45 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(3) = \int_{-1}^{3} g(t)\, dt = 45$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 10 and ± 18 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 18 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-10) - 18$ earns both points because numerical expressions need not be simplified.

· The $(-10) - 18$ expression earns the first point, but not the second point.

· The expression $-(-10) - 18 - 45$ earns neither point.

Note: Just the list $-10,\ 18$ earns neither point. However, if the values -10 and 18 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals $= -8$ earns just the second point because not enough work has been shown.

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0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] $\left| \int_{-4}^{-3} g(t)\ dt \right| = 10$ and $\left| \int_{-3}^{-1} g(t)\ dt \right| = 18$
- [ ] Answer

Solution:

$$h(-4) = \int_{-1}^{-4} g(t)\ dt = -\int_{-4}^{-1} g(t)\ dt$$

$$= -\int_{-4}^{-3} g(t)\ dt - \int_{-3}^{-1} g(t)\ dt = -(-10) - 18 = -8$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4,\ 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4,\ 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ 0)$, $(-1,\ 0)$, and $(3,\ 0)$, read the x -coordinates and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = -1$, and $x = 3$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points, if the response presents the connection between $g(x)$ and $h'(x)$ (that is, $g(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $g(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $g(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., " $h'(-3)$ changes from negative to positive") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

· $h(-4) = -8$

· $h(-3) = -18$

· $h(-1) = 0$

· $h(3) = 45$

· $h(4) = 40$

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Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

· At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.

· At $x = -1$, h has neither because $h' = g$ is positive for $x < -1$ and $h' = g$ is positive for $x > -1$.

· At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = -1$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-2, \ -1, \ 1$, and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· “ h' changes from increasing to decreasing or vice versa”

· “ h'' changes from positive to negative or vice versa”

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· " h' has a local extrema"

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x, it is eligible to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Location of inflection points
- [ ] Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that " h is differentiable $\Rightarrow$ h is continuous."

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since h is continuous, and $0 < 20 < 45$, such a point exists" would earn the point.

The value of 20 must be bounded by 0 and 45 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h\left(-1\right)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see $h\left(-1\right) = 0$.

A response may reference a value of $h\left(3\right)$ determined in part (a). If the value of $h\left(3\right)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 20.

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0	1

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-1 \le x \le 3. \Rightarrow h$ is continuous on $-1 \le x \le 3$.

$h(-1) = 0 < 20 < 45 = h(3)$

By the Intermediate Value Theorem, there must be a value of r, for $-1 < r < 3$, such that $h(r) = 20$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 45 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 45$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $0/0$ earns the first point.

· Indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $0/0$. For example, the statement $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{0}{0}$ does not earn the first point, but the statement $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 15x} \to \frac{0}{0}$ earns the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{2x}{h'(x) - 15}$, will earn the second point, but not the third point.

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Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x\to 3}\frac{x^2-9}{h(x)-15x}=\frac{0}{0}$ and $\lim_{x\to 3}\frac{x^2-9}{h(x)-15x}=\frac{2x}{h'(x)-15}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{2}{5}$, or its numeric equivalent.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h'=g$ is differentiable. $\Rightarrow h'=g$ is continuous.
$\Rightarrow h$ is continuous $\Rightarrow \lim_{x\to 3} h(x)=h(3)$ and $\lim_{x\to 3} h'(x)=h'(3)$.

$$\lim_{x\to 3}\left(x^2-9\right)=3^2-9=0$$

$$\lim_{x\to 3}\left(h(x)-15x\right)=h(3)-15\cdot 3=45-45=0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to 3}\frac{x^2-9}{h(x)-15x}=\lim_{x\to 3}\frac{2x}{h'(x)-15}$$

$$=\frac{2\cdot 3}{h'(3)-15}=\frac{6}{g(3)-15}=\frac{6}{0-15}=-\frac{2}{5}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x=-1$ is substituted for x , then $x=-1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$\cdot\, h'\left(x^2\right)\cdot 2x$

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$\cdot\, g\left(x^2\right) \cdot 2x$

$\cdot\, h'\left((-1)^2\right) \cdot 2 \cdot (-1)$

$\cdot\, g\left((-1)^2\right) \cdot 2 \cdot (-1)$

$\cdot\, h'(1) \cdot 2 \cdot (-1)$

$\cdot\, g(1) \cdot 2 \cdot (-1)$

$\cdot\, h'\left((-1)^2\right) \cdot (-2)$

$\cdot\, g\left((-1)^2\right) \cdot (-2)$

Responses that do not earn the point include:

$\cdot\, h'\left(x^2\right) \cdot 2$

$\cdot\, g\left(x^2\right) \cdot 2$

$\cdot\, h'(x) \cdot 2x$

$\cdot\, g(x) \cdot 2x$

$\cdot\, h'(1) \cdot (-2)$

$\cdot\, g(1) \cdot (-2)$

$\cdot\, h\left(x^2\right) \cdot 2x$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-1)$. For example, the second point is earned by $g\left(x^2\right)$ or by $g\left((-1)^2\right)$.

The second point may be earned without earning the first point. For example, $h'\left(x^2\right) \cdot 2 = g\left(x^2\right) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(1)$ to 19.7. The function g does not have to be stated explicitly. Responses that earn the first two points include:

$\cdot\, h'\left((-1)^2\right) \cdot 2 \cdot (-1) = 19.7 \cdot 2 \cdot (-1)$

$\cdot\, h'\left((-1)^2\right) \cdot (-2) = 19.7 \cdot (-2)$

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Recall, if $x = -1$ is substituted for x, $x = -1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g\left(x^2\right) \cdot 2x$ and $g\left((-1)^2\right) \cdot 2 \cdot (-1)$ earn the first two points.

The third point is earned for the correct answer of -39.4, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $h'(x)^2 \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$k'(x) = h'\left(x^2\right) \cdot 2x$

$= g\left(x^2\right) \cdot 2x$

$k'(-1) = g\left((-1)^2\right) \cdot 2 \cdot (-1) = g(1) \cdot (-2) = 19.7 \cdot (-2) = -39.4$

5-4b

38. Particle X moves along the positive x-axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$.

(a) Is particle X moving toward the left or toward the right at time $t = 1$? Give a reason for your answer.

(b) At what time $t \geq 0$ is particle X farthest to the left? Justify your answer.

(c) A second particle, Y, moves along the positive y-axis so that its position at time t is given by $y(t) = 7t + 3$. At any time t, $t \geq 0$, the origin and the positions of the particles X and Y are the vertices of a triangle in the first quadrant. Find the rate of change of the area of the triangle at time $t = 1$. Show the work that leads to your answer.

Part A

The response can earn up to 2 points:

1 point: For consideration of $x'(1)$

1 point: For correct answer with reason

$$x'(t) = 15t^2 - 18t$$
$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.

0	1	2

The response earns all of the following points:

1 point: For consideration of $x'(1)$

1 point: For correct answer with reason

$$x'(t) = 15t^2 - 18t$$
$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.

Part B

The response can earn up to 3 points:

1 point: For consideration of $x'(t) = 0$

1 point: For identification of $t = \frac{6}{5}$

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1 point: For correct answer with reason

$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0,\ t = \frac{6}{5}$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$.

0	1	2	3

The response earns all of the following points:

1 point: For consideration of $x'(t) = 0$

1 point: For identification of $t = \frac{6}{5}$

1 point: For correct answer with reason

$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0,\ t = \frac{6}{5}$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$.

Part C

The response can earn up to 4 points:

1 point: For the area function

$$\begin{aligned}\text{Area} = A(t) &= \tfrac{1}{2}x(t)y(t)\\ &= \tfrac{1}{2}\left(5t^3 - 9t^2 + 7\right)(7t + 3)\end{aligned}$$

2 points: For the derivative

$$A'(t) = \frac{1}{2}\left[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 6)(7)\right]$$

1 point: For the correct answer

$$A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$$

0	1	2	3	4

The response earns all of the following points:

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1 point: For the area function

$$\text{Area} = A(t) = \tfrac{1}{2}x(t)y(t)$$
$$= \tfrac{1}{2}\left(5t^3 - 9t^2 + 7\right)(7t + 3)$$

2 points: For the derivative

$$A'(t) = \frac{1}{2}\left[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 6)(7)\right]$$

1 point: For the correct answer

$$A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$$

39. Let h be a function with first derivative given by $h'(x) = x\cos\sqrt{x^3}$ for $x > 0$. At what values of x does h have local maxima on the interval $0 < x < 5$?

(A) 0.815 and 3.442

(B) 1.351 and 3.951 only ✓

(C) 2.811 and 4.945 only

(D) 1.351, 2.811, 3.951, and 4.945

Answer B

Correct. The local maxima of h occur at the values of x where the graph of h' changes from positive to negative. The graph of $h'(x) = x\cos\left(\sqrt{x^3}\right)$ on the interval $0 < x < 5$ has two zeros where the graph crosses the x-axis from positive to negative. These two zeros are at $x = 1.351$ and $x = 3.951$.

5-4b

40. The derivative of the function f is given by $f'(x) = 5(x-1)^2 \sin\left(x^2\right)$. On the interval $(1, 3)$, at which of the following values of x does f have a local minimum?

(A) 1.558

(B) 1.772

(C) 2.248

(D) 2.507 ✓

Answer D

Correct. The function f will have a local minimum where the graph of the derivative, f', changes from negative to positive. Graphing f' on the interval $[1, 3]$ shows that it changes from negative to positive only once. The calculator is used to find the zero of f' where this change occurs. That zero is at $x = 2.50663$.

41. If $f'(x) = 1 + 2x - x^3 - \sqrt{x^4 + 1}$, then f has a local minimum at $x =$

(A) -1.836

(B) -1.117

(C) 0 ✓

(D) 1.181

Answer C

Correct. The function f will have a local minimum at a critical point where the graph of f' changes from negative to positive. Graphing f' on the calculator reveals only one critical point where f' changes from negative to positive, which is at $x = 0$.